Characterizing Canadian Out-of-district Political Donations from 2015 to 2024*

Benedict Cummins-Mburu   Rohan Alexander

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Federal representation in Canada is structurally geographical: one Member of Parliament (MP) represents one electoral district. Despite this, Canadians can legally donate to a number of political entities, regardless of proximity. Using Elections Canada's Open Data on Contributions matched with socio-demographic data from the Canadian Census, we characterize broad patterns in Canadian out-of-district donation from 2015 to 2024. We find that out-of-district donations form a persistent share of donor-sourced funding to both candidates and district associations, averaging 40% across time, election periods, and parties. While relatively constant through time, the propensity for out-of-district donation varied substantially across districts, and more than what we would expect from their spatial arrangement alone. Districts with higher population density, median household income, educational attainment, and visible minority proportions see higher rates of out-of-district giving, the same traits that predict donor prevalence generally. Still, donor-sourced local campaign funding remains largely regionalized; nearly half of donations that leave their district are received by a political entity in a neighbouring one, and few cross provincial lines. These and other findings may inform further research into the geography of local campaign finance, in Canada and other Westminster-style parliamentary democracies.

Keywords: out-of-district donation; Canadian politics; political geography; political financing

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1 Introduction

In Canada, federal representation is structurally geographical. Citizens are grouped into a few hundred electoral districts based on where they live, each represented by a single Member of Parliament (MP). An individual's political influence in this system is canonically exercised by voting for their candidate of choice during election periods. Beyond the ballot, alternate forms of political engagement exist to support a broader scope of activities, entities, and without the spatio-temporal constraints imposed by elections. Citizens may volunteer at their local district associations, advocate for their preferred national party online, or contribute financially to a candidate's campaign (Baker 2016; Garnett 2024).

Whether intentional or not, these alternative forms of political participation may serve to extend a citizen's political influence beyond their district's boundaries, and among these, donations are perhaps the clearest and most flexible mechanism of doing so (Gimpel et al. 2008). In Canada, citizens and permanent residents can legally contribute to a host of federal political entities at any time, in any district, and as often as they want, subject to annual and per-election limitations (Elections Canada 2026a). Beyond their flexibility, individual donations are a major source of funding for individual constituency campaigns (Cross et al. 2020; Garnett 2024; Elections Canada 2026d). While campaign expenses may be partially reimbursed through public subsidies, most funds needed for a candidate to carry out an effective campaign must be available on-demand (Garnett 2024). National parties do transfer funds to their candidates, but a district's local apparatus – the candidate and their association – is routinely expected to generate the bulk of its own funds from individual donors.

Despite a growing body of Canadian research documenting the extent of, and patterns in, federal political donations generally (e.g. Besco and Tolley 2022; Tolley et al. 2022; Garnett 2024), comparatively little attention has been given to out-of-district donations. Existing research in the U.S. documents substantial rates of out-of-district giving during congressional campaigns, wherein the typical House (Senate) candidate sources over two-thirds (half) of their campaign donations from out-of-district (out-of-state) (Gimpel et al. 2008; Sievert and Mathiasen 2023). These donations are thought to be primarily motivated by partisan gain (Gimpel et al. 2008; Jacobs and Imboywa 2024), and redirect legislators' responsiveness from local constituents toward national donor bases (Baker 2016; Canes-Wrone and Miller 2022; Jacobs and Imboywa 2024; Nathan et al. 2025). Further, and to an overwhelming degree, these donations originate from a single small group of disproportionately wealthy, older, white, and well educated districts (Gimpel et al. 2006; Bramlett et al. 2011).

In this paper, we document the extent of, and large-scale trends in, Canadian out-of-district donations to electoral candidates and district associations from 2015 to 2024. We use a database containing all contributions received by federal political entities during this period, made publicly available by Elections Canada and pre-processed by the Investigative Journalism Foundation. This dataset was augmented by situating donors geographically, and matching donations to their socio-demographic environment using the 2021 Census, allowing us to better understand which factors correlate with out-of-district donation rates. In addition to our findings, this study contributes a cleaned, de-duplicated, and analysis-ready dataset of individual contributions to all federal political entities from 2015 to 2024, which we have made available upon request[1].

We find that, between 2015 and 2024, out-of-district donations account for a persistent share of donor-sourced funding to both electoral candidates and district associations, averaging roughly 40% across time, election periods, and party lines. This relative invariance does not extend to space, however. The propensity for out-of-district donation changes substantially from district to district. Donations originating from densely populated, high-income, and well-educated districts are more likely to leave their district of origin, echoing predictors of out-of-district giving established in the U.S. Our analysis also shows that the visible minority proportion of a donor's district was strongly correlated with out-of-district propensity. This departs from historically white "donor districts" of the U.S. (Gimpel et al. 2006; Bramlett et al. 2011; Canes-Wrone and Miller 2022), but aligns with recent Canadian studies linking minority ethnic communities to higher rates of in- and out-of-district giving (Garnett et al. 2022; Besco and Tolley 2022). Despite the prevalence of out-of-district donations, direct donor funding remains regional: nearly half of all donations that leave their district of origin are addressed to political entities in a neighbouring one, and less than one-fifth crossed a provincial boundary, limiting their potential influence on geographically based representation.

Although we do not formally test the representational impacts of out-of-district donations, their systematic characterization presented here provides a foundation for future research on this topic, and improves understanding of the origins of local campaign finance.

2 Background

2.1 Out-of-district Donation in the U.S. and its Representational Effects

Our study is motivated by existing research on out-of-district donations to U.S. members of Congress (representatives and senators). There, members of both chambers formally represent, and are elected by citizens living in geographically defined regions of the country (districts and states, respectively). Like in Canada, individual constituency campaigns are funded in large part by individual donors (Ansolabehere et al. 2003; Corrado et al. 2010; Barber 2016), which need not reside in their constituency. U.S. scholars describe big donors and their environments as wealthy, older, white, and well-educated (Francia et al. 2003; Mandle 2007; Gimpel et al. 2006), characteristics which are decidedly unrepresentative of the typical U.S. voter (Bonica et al. 2013; Hill and Huber 2017; Grumbach and Sahn 2020). Given this, and broader trends in the "nationalization" of campaign finance (Hopkins 2018; Sievert and Mathiasen 2023; Jacobs

[1]The dataset cannot be redistributed directly, but it can be reconstructed in full by following the steps at https://zenodo.org/records/21921990. Doing so requires a file licensed from Canada Post (Statistics Canada 2025), as well as a pre-processed dataset obtained on request from the Investigative Journalism Foundation.

and Imboywa 2024), recent U.S. literature has sought to (1) document rates of out-of-district donation, and (2) test whether such donations warp the classically “dyadic” (Weissberg 1978) member–constituent relationship in favour of “surrogate” (Mansbridge 2003) representatives; legislators more responsive to their national donor base.

Research describing nationalized donations and donor characteristics during electoral campaigns in the U.S. is not new (Francia et al. 2003; Mandle 2007; Gimpel et al. 2006). However, it can be argued that Gimpel et al. (2008)’s examination of out-of-district donations during four Congressional elections between 1994 and 2004 catalyzed research on out-of-district donations and their representational effects. During this period, the typical candidate received roughly two-thirds of individual donations from out-of-district, a rate reproduced for more recent elections by Canes-Wrone and Miller (2022). For both Republican and Democrat candidates, a large majority of these donations originated from a shared set of so-called “donor districts”: a small number of wealthy, educated, older, and affluent districts. Notably, these socio-demographic predictors are largely identical to those predicting a region’s donor output in general (Gimpel et al. 2006). Because district competitiveness strongly predicts the share of donor funding received from out-of-district, they argue that this behaviour is primarily motivated by partisan advantage. More recent studies on out-of-state donations to Senate candidates report out-of-state donation shares over 50%, which have increased steadily over the century, for both parties (Jacobs and Imboywa 2024; Sievert and Mathiasen 2023; Keena 2024).

Beyond descriptions of the phenomenon, U.S. literature on out-of-district donations offers strong evidence that these contributions have tangible representational effects. Bramlett et al. (2011) pair national donation and survey data to show that donors from Gimpel et al. (2008)’s “donor districts” have a unique political ecology distinct from other parts of the country, which likely intensifies bias in opinion. Baker (2016) study the 2006-2010 Congressional elections, revealing that representatives become less responsive to their constituencies’ ideological preferences when heavily reliant on out-of-district contributions. Canes-Wrone and Miller (2022) further explore the 2006-2016 Congressional elections to show that House voting behaviour is indeed associated with the preferences of their national donor base, contrasted against earlier research finding that political action committees (PACs) do not affect roll call behaviour (Ansolabehere et al. 2003). Studies on Senatorial elections also argue that out-of-state-donations have meaningful representational impacts (Jacobs and Imboywa 2024). For instance, Nathan et al. (2025) experimentally demonstrate that constituents who learned about outside donation expected their senators to be less loyal to local interests, emphasizing the rivalrous relationship that may form between local constituents and national donor bases.

2.2 Individual Donations and Campaign Financing in Canada

Few studies analyze out-of-district donation shares to Canadian parliamentary candidates[2], or electoral district associations (EDAs). We propose three potential reasons why. Most obviously, U.S. campaign financing moves vastly more money, and sustains a larger research community which has recently been interested in the nationalization of campaign finance. Second, location-disclosed donations in the U.S. (those whose donors gave over $200) account for an overwhelming majority of individual donor funding (Corrado et al. 2010). While Canadian donors who have contributed over $200 must also disclose their location (Elections Canada 2026d), contribution limits are stricter than in the U.S., so smaller, location-anonymized donations make up a larger share of donor-based funding to federal political entities overall. Third, Canadian contributions records are not presented in an analysis-ready format: the researcher must situate donors within districts and dissemination areas manually, and records are often duplicated since big databases often merge Annual, Quarterly, and Electoral financial returns (Elections Canada 2025a).

This lack of research is not, however, a signal that direct individual donations are an unimportant piece of Canadian campaign financing. District campaigns are run by candidates, with financial support from their electoral district associations[3]. Together, these entities receive the majority[4] of their combined funding from three[5] sources: individual contributions (limited to $1775 per contributor[6]), unlimited transfers from their affiliated central party, and unlimited transfers to the EDA from other EDAs (Cross et al. 2020; *Canada Elections Act* 2000). While in principle this funding scheme allows local entities to source most of its funds directly from the central party, in practice, this is rarely the case. For instance, during the 2015 general election period, central party transfers supplied roughly one-quarter of campaign funding[7], and inter-EDA transfers less than ten percent (Cross et al. 2020). Local campaign reliance on individual funds is consistent with the established “stratarchical” organization of Canadian parties and party finance (Carty 2002; Coletto et al. 2011; Cross et al. 2020). While national parties exert undeniable influence over their local district campaigns, scholars note that Canadian parties have decentralized candidate selection processes, and a high tolerance for personalized campaigns tackling local issues (Cross and Young 2015).

[2]A notable counter-example is Besco and Tolley (2022), discussed later in this section.

[3]Non-partisan (i.e. independent) candidates do not have EDAs, and are therefore funded up-front entirely by individual donors.

[4]We are assuming that other legal transfers (e.g. transfers from leadership contestants, nomination contestants) are negligible for the average district association or candidate.

[5]Candidates that received at least 10% of votes are eligible to have 60% of their election expenses reimbursed through public subsidies (Elections Canada 2026c). While an important source of funding, we feel that these do not undermine the importance of individual contributions, since they are not available up-front.

[6]This limit applies on an annual basis for district associations, and a per-election basis for candidates (Elections Canada 2026d). $1775 CAD is accurate as of 2026, and increases by $25 CAD every year.

[7]Similar proportions were found in Coletto et al. (2011)’s study of campaign finance. In both studies, campaigns from all three major Canadian parties sourced, at most, 30% of their funding from the central party.

Existing research on Canadian donations has described who donates to electoral candidates[8]. Initial analyses of individual donations from 2006 to 2011 (Jansen et al. 2012; Carmichael and Howe 2014), and a more recent analysis of candidate-bound donations during the 2015, 2019, and 2021 general elections (Garnett 2024) have found that donors are more likely to originate from older, wealthier, and better-educated districts. Garnett (2024) in particular finds also that donations are more likely to originate from regions with higher visible minority proportions. Survey-based studies by Tolley et al. (2022) and McMahon et al. (2023) find that women are less likely to donate than men. These socio-demographic characteristics are similar to those attributed to U.S. donors, with the exception of ethnicity.

Related studies suggest that Canadian donors are more likely to donate to candidates with whom they share aspects of their identity with. For instance, women donors are more likely to donate to women candidates (Tolley et al. 2022), and rates of co-ethnic donation are high, particularly among South Asian donors (Besco and Tolley 2022). Another study analyzing donor-candidate co-ethnic affinity reports that the likelihood of a donor donating out-of-district is higher for ethnic minorities (50 - 70%) compared to donors of European origin (33%) (Besco and Tolley 2022).

3 Data and Methods

3.1 Overview of Data Acquisition and Processing

The data used in this paper was originally sourced from the Elections Canada Open Data on Contributions datasets (Elections Canada 2025a). Since 2004, the *Canada Elections Act* has required that all individual contributions to federal political entities[9] valued at over $20 be recorded by the recipient, shared with the government, and made publicly accessible (Elections Canada 2026d). In addition, donors whose cumulative sum of contributions exceeds $200 are required to disclose their full name and address, including their postal code. For all contributions, Elections Canada also records the receiving political entity's name, district affiliation (if applicable), party (if applicable), type (e.g. national party, candidate, etc.), and the date the contribution was received. The primary dataset used in this study is a pre-processed version of Elections Canada's, made available to us by the Investigative Journalism Foundation. Their cleaning process is described in Appendix A.

To even be considered out-of-district, a federal donation must be received by a political entity that is representationally tied to a specific district. For this reason, our study primarily focuses

[8]We refer the reader to Young and Jansen (2011), Garnett et al. (2022), Pruysers (2024), and Mockler and Garnett (2024) for Canadian-context studies on contributions to other political entities.

[9]Donations (monetary or otherwise), loans, and loan guarantees received by national parties, EDAs, leadership contestants, nomination contestants, and electoral candidates are covered by the *Act*.

on donations sent to candidates (during election periods) and district associations[10], hereafter referred to as localized entities. However, given the prevalence of donations sent to national entities (parties or leadership contestants) in our dataset, we briefly compare them with those addressed to localized entities in Section 3.2.

Every ten years, federal electoral district (FED) boundaries are redrawn to account for population changes that occurred within the decade (Elections Canada 2025b). To restrict our focus to a single, static set of FEDs, our analysis only includes contributions made during the 2013 representational order, effective between October 2, 2015 and April 22, 2024 (Elections Canada 2025b)[11]. This is the largest order present in the dataset, capturing 61% of all recorded contributions since 2004, and spanning three general election periods (42nd, 43rd, and 44th). Donors were matched to their FEDs by their postal codes using Statistics Canada's Postal Code Conversion Files[12] (Statistics Canada 2025), made available to us by the University of Toronto. Donor FEDs were then compared with recipient FEDs to assess whether a donation was out-of-district. Donor location was not recorded for the majority of donations under \$200, and the small share that could be geolocated were distributed very differently from the rest of the data. For these reasons, this study only considers donations above \$200, as was done in similar studies (Gimpel et al. 2008; Garnett 2024). These smaller donations are discussed further in Appendix C.

Postal codes were also used to situate donors within census dissemination areas (DA), allowing us to attribute socio-demographic characteristics to their environments at a fine spatial grain. For each dissemination area, we identified its population density, median household income, mean age, proportion of the population with a post-secondary education, and proportions of various visible minority ethnicities. A full list of the ethnicities we consider is available in Appendix B. These variables were informed by similar studies on donor demography (Gimpel et al. 2006, 2008; Garnett 2024). Further, ethnicities were disaggregated since previous Canadian research has found that some minority groups such as South Asians donate at much higher rates than others (Besco and Tolley 2022). We again used Postal Code Conversion Files to match donor postal codes to the DAs of the 2021 Census, and to group DAs into FEDs[13] to deduce the same socio-demographic information at the district level.

As a final step, each district was assigned a competitiveness score. This was calculated as the average margin of victory of elected MPs across the 42nd, 43rd, and 44th general elections. Data on electoral outcomes was sourced from Open Government Canada (Treasury Board

[10]Although nomination contestants and by-election candidates are also local political entities and were present in the data, these were not considered to simplify our analysis and discussion, since they only make up 0.7% of donations addressed to localized entities.

[11]This date range successfully excluded donations whose recipients were associated with other representational orders. Only 0.5% of donations with recipients associated with the 2013 representation order were missed.

[12]Postal codes may straddle multiple FEDs. For this study, we created a deterministic spatial lookup table that resolves these cases using reasonable criteria. Donations whose postal codes could not be matched in this way were dropped. Our methodologies, and loss impact assessments are detailed further in Appendix B.1.

[13]Dissemination areas may straddle multiple FEDs. Appendix B.1 discusses how these cases were handled.

of Canada Secretariat 2026), and its use in our analysis is detailed in further sections. A detailed walkthrough of the full data pipeline, from the files available for download on Canadian government websites to the core datasets used for analysis can be found in Appendix A and Appendix B. All data cleaning, validation, visualization, and analysis were conducted using the R statistical software (R Core Team 2025).

3.2 Data Summaries

3.2.1 Situating Localized Donations

We first provide summaries of all donations[14] during our study period, to help situate further analysis of out-of-district donations to localized entities. Table 1 breaks down donations by recipient type. Parties and leadership contestants received the majority (80.8%) of contributions during our study period. Among donations addressed to localized entities, 24.3% are received by candidates, and 75.7% were received by district associations (EDAs). During election periods, donations to candidates formed 48% of the funding received by localized entities, with district associations making up the remaining 52% (see Appendix F).

Figure 1 shows how contributions during our study period were distributed in time[15], for national and localized entities separately. Among national-level donations, we observe three distinct temporal signatures. First, donation rates are higher in December of every year during our study, likely since charitable donations tend to be more common in this period in general, as individuals claim tax deductions and give more during the holidays. Second, we observe six peaks corresponding to general election periods (peaks during and immediately before grey shaded regions), and Conservative party leadership campaigns (peaks in the blue trend line in the first half of 2017, 2020, and 2022). Third, national-entity donation rates generally increased over our study period, following trends observed from 2004 to 2015 by McMahon and Wolfe (2015), and particularly following the 2021 general election in the Conservative party.

Donation rates to localized entities exhibited the same December and election period peaks that were observed for donations to national entities, however an increasing trend was not observed. This agrees with trends observed in candidate-bound donations by Garnett (2024) and Elections Canada (2026b) over similar periods. We also observe a significant reduction in donations to EDAs from the start of 2019 to mid-2021. This may in part be attributed to the COVID-19 pandemic (McMahon et al. 2023), however it is not entirely clear why donations to district associations do not appear in our dataset during this period. Simple spatial comparisons

[14]From Section 3 onwards, the terms "donation" and "contribution" are used interchangeably and implicitly refer to individual contributions above $200 that were successfully donor-geolocated.

[15]All donations to electoral candidates and leadership contestants were tied to specific events in time. A small proportion (0.7% of the total contributed amount) of contributions to EDAs and national parties over $200 were missing specific dates, but could always be attributed to a fiscal year. Plots over time in this report include these cases, which slightly exaggerates the observed December peaks in contribution rates.

Table 1: Breakdown of contributions over $200 to major Canadian federal political entities, from 2015 to 2024. We note that by-election nomination contestants and by-electoral candidates are excluded from this Table.

Political Entity	Type	Donation Count	Donation Amount (Thousands)	Share of Total Amount
Candidate	Localized	30,559	$19,883	4.6%
District Association	Localized	108,619	$63,354	14.6%
Leadership Contestant	National	47,873	$32,451	7.5%
Party	National	618,609	$319,056	73.4%

between the donor bases of national and localized entities during our study period yielded only minor differences, and are presented in Appendix F.

3.2.2 Spatio-temporal Trends in Out-of-district Donation Rates

In this section, we present data summaries describing when and where out-of-district contributions are more or less common. Table 2 displays the proportion of donations above $200 (raw and amount-weighted) that were sent out-of-district, for donations to candidates and EDAs. Overall, candidates received 37.6% of their donations from outside their district, and EDAs received 43.8%. In both cases, these proportions increase when donations are weighed by their value, indicating that out-of-district donations are larger, on average, than their within-district counterparts.

Table 3 and Figure 2[16] show how out-of-district funding rates change over time, for candidates and district associations respectively. Rates for candidates show a slight decreasing trend, from 39.5% in the 42nd general election in 2015 to around 35.4% in subsequent periods (Table 3). However, this may reflect the heightened absolute number of contributions received in 2015, due to that election's unusually long writ-period. EDAs received roughly the same proportion of out-of-district contributions during the entire study period, including during writ periods[17].

Figure 3 displays the relationship between in- and out-of-district contribution amounts, for each electoral district present in our dataset. We display donations to candidates and district

[16]We emphasize the trend instead of raw shares to control for large month-to-month variation in donation rates to EDAs (shown in Figure 1).

[17]Roughly similar rates, and the same consistency through time, were observed when entities from individual parties were examined (Appendix F).

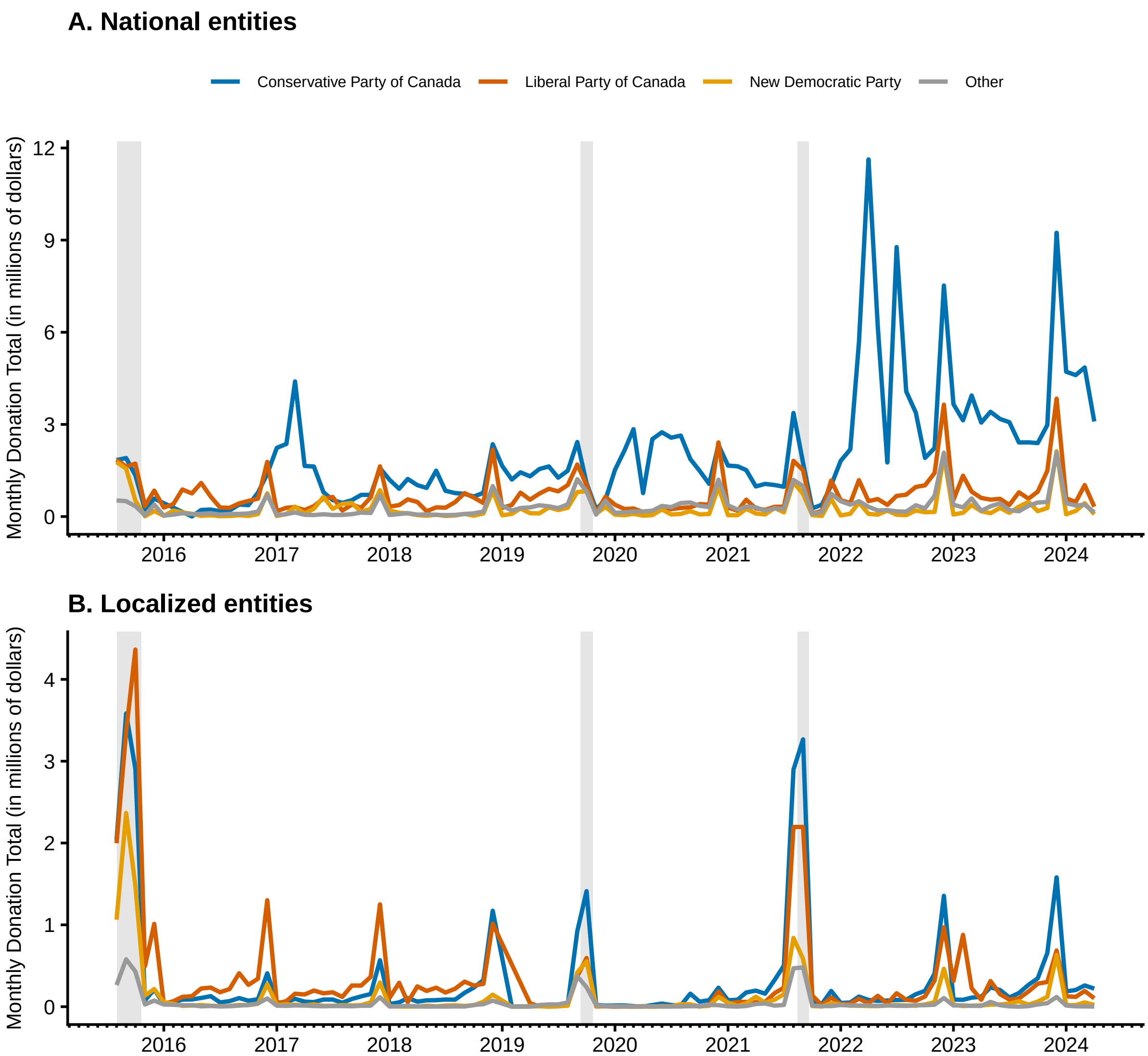


Figure 1: Time series of contributions above \$200 to national (A) and localized (B) federal political entities from 2015 to 2024. Within each panel, trendlines are separated by the party of the recipient. Shaded vertical bars indicate the 42nd, 43rd, and 44th general election periods respectively. Series were qualitatively similar when contributions under \$200 were included.

Table 2: Proportion of contributions over \$200 to candidates (excluding by-elections) and district associations that were received from outside their district, from 2015 to 2024. The number and total value of contributions received by these entities is reported for reference.

Political Entity	Donation Count	Donation Amount (Thousands)	OOD Proportion	OOD Proportion (Amount Weighted)
Candidate	30,559	\$19,883	37.6%	40.2%
District Association	108,619	\$63,354	43.8%	47.8%

Table 3: Proportion of contributions to candidates (excluding by-elections) over \$200 that were out-of-district (OOD) over the three general election periods that occurred during our study period. Raw and value-weighted proportions are reported, along with the overall number of contributions (in- and out-of-district) for reference.

General Election	Donation Count	Donation Amount (Thousands)	OOD Proportion	OOD Proportion (Amount Weighted)
42nd general election	16,662	\$10,383	39.5%	41.9%
43rd general election	7,900	\$5,204	35.2%	39.1%
44th general election	5,997	\$4,296	35.6%	37.7%

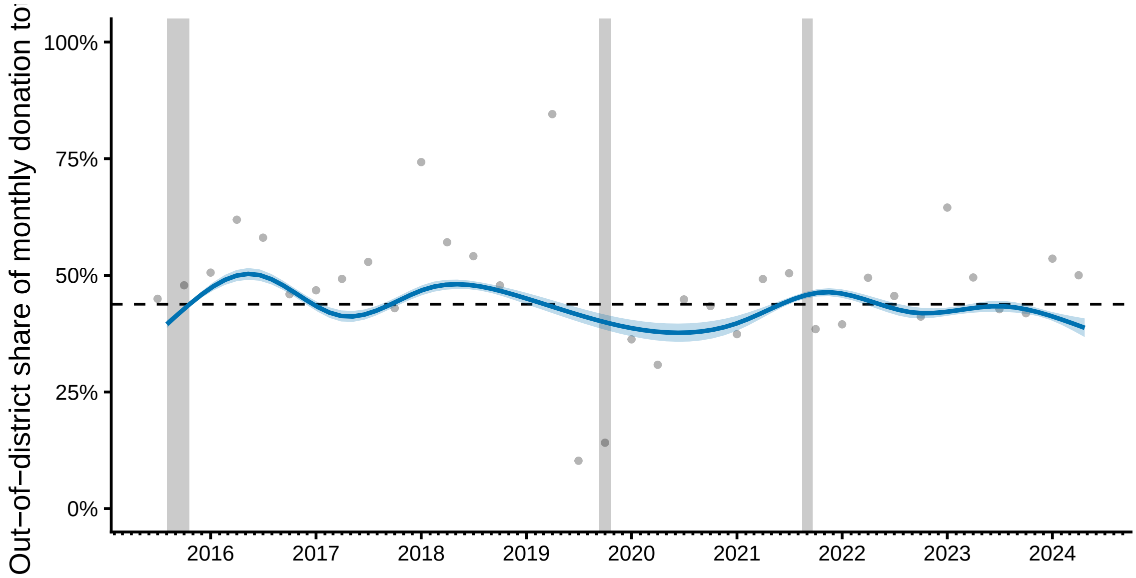


Figure 2: Time series showing the monthly share of donations (over $200) to district associations received from out-of-district donors, from 2015 to 2024. Raw monthly shares are shown as grey points, and the fitted trend over time (with its 95% confidence interval) is shown with the blue line and ribbon. Trend was fit using a generalized additive model with a smooth term over donation date, estimated at the individual level. Shaded vertical bars indicate the 42nd, 43rd, and 44th general election periods respectively. The horizontal dashed line indicates the overall out-of-district share, for reference.

associations separately. In both cases, out-of-district contribution rates vary greatly from district to district[18]. Some Toronto metropolitan districts such as Toronto—St. Paul's and University—Rosedale sent between 70% and 80% of their contributions to localized entities in other districts, while others such as Alberta's Lethbridge and Ontario's Niagara Falls only sent between 5% and 15% of their localized contribution sum to other districts. Many districts, especially for donations addressed to candidates, have low out-of-district donation rates, as is illustrated by the line of points near $y = 0$ in Figure 3. We also find that, at least among the few top districts by contribution amount, out-of-district propensity is not a simple function of urbanization[19]; Vancouver Granville in British Columbia and Brampton North in Ontario are similarly metropolitan, but have very different out-of-district contribution rates to electoral candidates. District-level differences in candidate-bound out-of-district contribution rates are even more pronounced when election periods are examined individually (see Appendix F).

3.2.3 Spatial Reach of Out-of-district Donations

Our final set of data summaries focus on the out-of-district donations themselves, examining their reach. Table 4 shows the proportion of out-of-district contributions bound for neighbouring districts, and districts in other provinces or territories. For both candidates and EDAs, around 40% of out-of-district donations are sent to adjacent districts, and only around 12% reach other provinces. Out-of-district donations are overall closer to home for candidate-bound donations, echoing our findings in Table 2. For the interested reader, we provide a visualization of inter-provincial donor funding networks in Appendix F.

The distribution of distances travelled by out-of-district donations[20] is shown in Figure 4. We observe that most out-of-district donations travel around as far as the average distance between two metropolitan districts (green vertical line). Expectedly, the likelihood of a donation travelling some target distance declines steeply as that target distance increases, becoming extremely rare past 1000km. Two exceptions to this rule are donations which travelled around 2700 and 3400 kilometres (orange and blue lines in Figure 4) respectively, which roughly demarcate the distances between Toronto and Calgary, and between Toronto and Vancouver. Overall, these summaries suggest that out-of-district donations are largely regionalized, and, for the most part, contained within their province of origin. Those few donations that breach

[18] The marginal distribution of out-of-district proportions across all districts is shown in Appendix F.

[19] Since districts vary enormously in size and shape depending on their population density, one might naturally expect that tightly-packed urban districts would have higher out-of-district contribution rates, simply due to the spatial nature of socio-political interest. In Appendix E, we simulate the out-of-district share for each district under this expectation, and find that district-level out-of-district proportions differ much more than what can be explained by the spatial arrangement of districts alone.

[20] The "distance travelled" by an out-of-district distribution was taken to be the Haversine great-circle distance between the centroid coordinates of the donor's dissemination area and the centroid coordinates of the recipient's district.

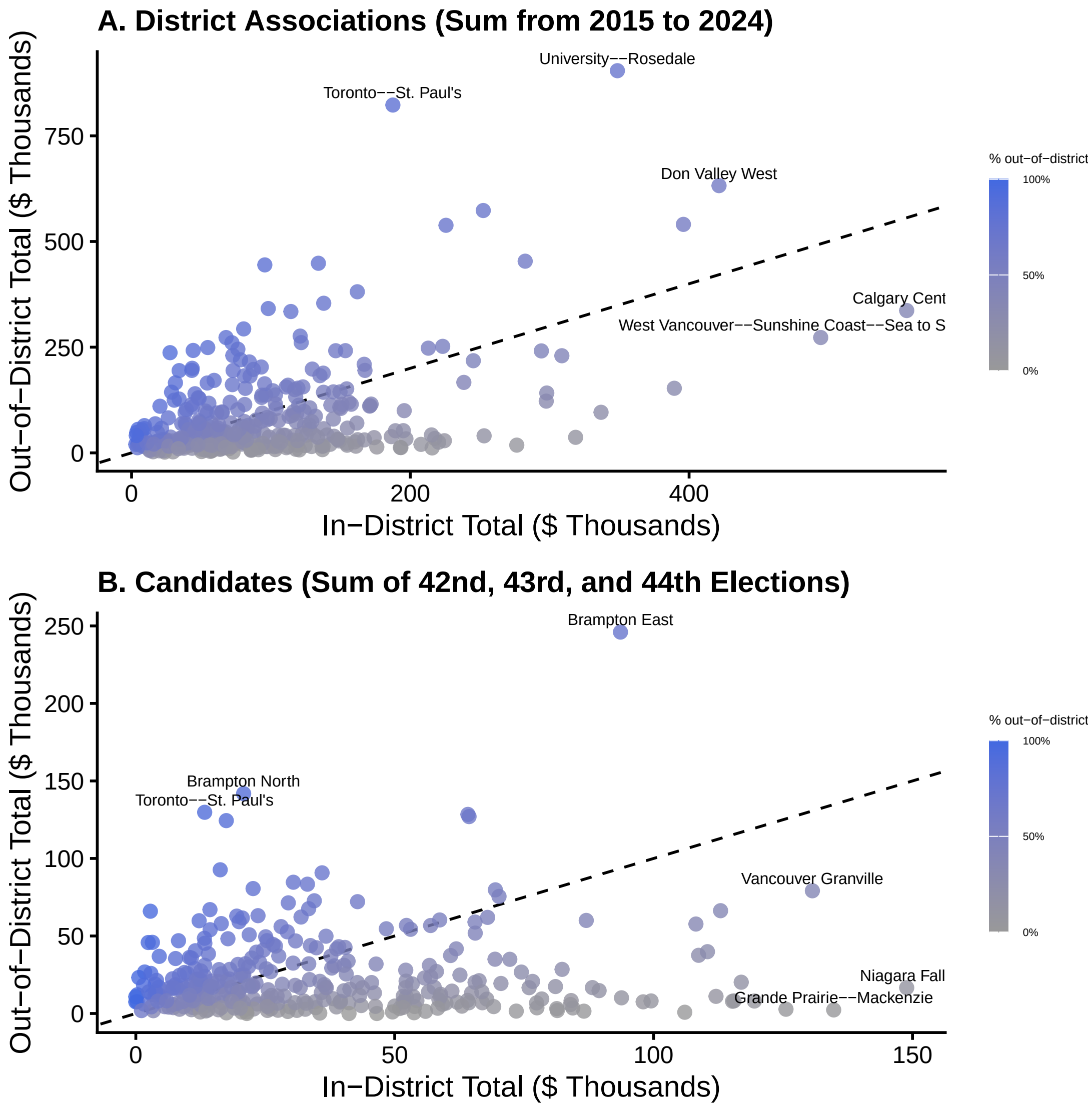


Figure 3: Relationship between the amount donors of a given district sent in- v.s. out-of-district, from 2015 to 2024. Points represent individual districts, and are colored based on the share of funds sent to (A) district associations or (B) candidates in other districts. The top 3 districts by amount contributed in- and out-of-district respectively are labelled. Graphs are similar when donor counts are displayed instead of aggregated donation amount. Subplot B looks similar when specific election cycles are considered. The grey dashed line is 1:1.

Table 4: Proportion of out-of-district contributions over $200 that were sent to adjacent districts, and out-of-province during our study period. Raw and value-weighted proportions are reported, along with the overall number of contributions for reference. Contributions received by candidates outside general election cycles (by-elections) are not included in calculations.

Received by	Sent to	By count	By amount
Candidate	Adjacent district	43.3%	44.1%
Candidate	Out-of-province	10.6%	9.7%
District association	Adjacent district	37.1%	36.0%
District association	Out-of-province	14.7%	14.4%

provincial boundaries are most likely dominated by donations between the metropolitan centers of each province.

3.3 Out-of-district Propensity Models

3.3.1 Individual-Level Model

We used a logistic regression model to explore how the likelihood of out-of-district donation varies across different transactional characteristics, donor demographic environments, and district competitiveness, for individual donations. Our outcome variable is binary: whether or not a donation was sent out-of-district. This model was fit on both donations to EDAs and candidates combined. Models fit on each entity separately had similar coefficient magnitudes, signs and significances (see Appendix D). Donations were not grouped by individual, since we were interested in both donor and recipient characteristics in this model. Diagnostics showed that residual autocorrelation, the primary statistical risk associated with this decision, was not a large concern in the model.

The transactional predictors in our model are: the type of recipient (candidate or district association), their political party, and the monetary value of the donation. We did not include time-indexed variables since our data summaries showed that out-of-district rates were constant over time, for both candidates and district associations. Following Garnett (2024), we also considered the following characteristics of the donor's socio-demographic environment: province (grouped into regions), mean age, median household income, the share of the population with a post-secondary degree, and the share of the population that is a visible minority[21]. These predictors are associated with out-of-district donation in the U.S. (Gimpel et al. 2006, 2008). These environmental characteristics were assigned to donations based on their donor's

[21]An alternative model where regions were disaggregated into provinces showed qualitatively similar results. An alternative model where the visible minority proportion was disaggregated into specific ethnic groups is shown in Appendix D.

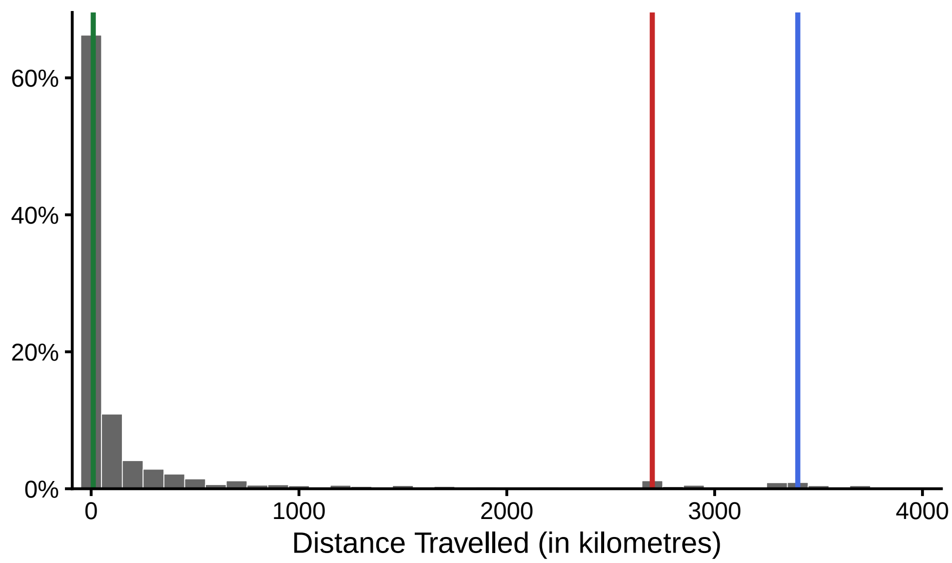


Figure 4: Distribution of distance travelled, for out-of-district donations over $200 during our study period. Both donations to candidates and district associations are shown together, however distributions were very similar for both entities separately. Distance travelled is taken to be the distance between the centroid of the donor's dissemination area and the centroid of the recipient's district. Green, red, and blue vertical lines respectively mark 10, 2700, and 3400 kilometres. Distribution shown is amount-weighted, however a count-weighted distribution was similar.

dissemination area. District competitiveness was proxied using the difference between the percentage of votes received by the winning and runner-up candidates, averaged across all three general elections captured in our study period[22], inspired by Garnett (2024). This predictor was included since we might expect donors from less competitive districts to donate out-of-district more often, recognizing that a local contribution will do little to affect an outcome that is effectively predetermined.

We also included the population density of the donor's district, for two reasons. First, out-of-district donors are more likely to come from urban (and therefore more densely populated) districts in the U.S. (Gimpel et al. 2008). Second, population density serves as a relatively good proxy for how spatially clustered a district is, since Canadian districts are designed to have similar population sizes (Elections Canada 2025b). Out-of-district donation rates are likely naturally higher in smaller districts, since donors are more likely to interact with individuals, businesses, or politicians close by that happen to be located in a different district. Population density was transformed in the model by applying a square root, allowing us to interpret the metric as the inverse of the average distance between people, which captures this distance effect more faithfully[23].

The resulting model is:

$$\begin{aligned}\text{logit}(p_i) = \beta_0 &+ \beta_1 \text{Election}_i + \beta_2 \text{Candidate}_i + \sum_{j=3}^{5} \beta_j \text{Party}_{ji} + \beta_6 \text{Amount}_i \\ &+ \sum_{k=7}^{11} \beta_k \text{Region}_{ki} + \beta_{12} \text{Age}_i + \beta_{13} \text{Income}_i + \beta_{14} \text{Education}_i + \beta_{15} \text{VisMin}_i \\ &+ \beta_{16} \text{VictoryMargin}_i + \beta_{17} \text{Density}_i\end{aligned}$$

where:

- p_i is the probability that donation i was an out-of-district donation.
- Election_i indicates whether donation i occurred during a general election period.
- Candidate_i indicates whether donation i was addressed to a Candidate (baseline is EDA).
- Party_{ji} indicates which party (among Liberal, N.D.P, or Other, with baseline being Conservative) the recipient is associated with.
- Amount_i is the monetary value of donation i.
- Region_{ki} indicates which region (among Atlantic Canada, B.C., Prairies, Quebec, Territories, with baseline being Ontario) the donor is located.

[22] Districts were more likely to have elected an MP from the same party (or simply the same MP) for all three elections when their average margin of victory is higher (pseudo$R^2 = 0.22$), meaning that this competitiveness metric also captures a decent share of the variation in district-level party stability.

[23] The observed spatial variation in out-of-district propensity cannot be fully explained by the spatial arrangement of districts alone (Appendix E), however it remains an important consideration.

- Age_i is the average age of residents in the donor's dissemination area.
- Income_i is the median household income in the donor's dissemination area.
- Education_i is the proportion of residents in the donor's home district holding a post-secondary qualification.
- VisMin_i is the proportion of residents in the donor's home district who identify as a visible minority.
- VictoryMargin_i is the difference between the percentage of votes received by the winning and runner-up candidates, averaged across all three general elections in the donor's home district.
- Density_i is the population density (persons per km^2) of the donor's home district. As mentioned in-text, the model was fit using the square root of this variable.

3.3.2 District-Level Model

Inspired by the large variation in out-of-district giving between districts observed in Figure 3, we fit a second regression model to explore which socio-demographic features were associated with higher rates of out-of-district donation, aggregated at the district level. Our outcome variable in this case was the proportion of the total contributed amount[24] sent out-of-district during our entire study period, for each district. Given that this outcome is a proportion, we used beta regression, fit using the `betareg` package in R (Cribari-Neto and Zeileis 2010). This model was fit on both donations to EDAs and candidates combined, following from the individual model. A model fit on only EDAs separately had similar coefficient magnitudes, and the same signs and significances (shown in Appendix D). A model fit on only candidates was not fit due to data sparsity and the presence of singularities in the response.

The covariates included in this model were the same environment variables chosen for the individual-level model, aggregated at the district level[25]. We also included the same district-level competitiveness metric used in the individual model. Since districts had very different numbers of donations used to calculate our outcome proportion, we also assumed a power-law relationship between the common precision parameter ϕ and the number of donations, following Simas et al. (2010).

The resulting model is:

[24] We observed high correlation ($\rho = 0.978$) between the proportion of out-of-district donations and the share of total contributed amount that left the district, for both candidates and EDAs. Therefore, results would have been similar if we had used the number of donations as our outcome instead of the amount-weighted share of donations.

[25] As with the individual-level model, an alternative model specification with disaggregated regions and visible minority proportion is shown in Appendix D.

$$
\begin{aligned}
p_i &\sim \text{Beta}(\mu_i, \phi_i) \\
\text{logit}(\mu_i) &= \beta_0 + \beta_1 \text{Amount}_i + \sum_{j=2}^{6} \beta_j \text{Region}_{ji} + \beta_7 \text{Age}_i + \beta_8 \text{Income}_i \\
&\quad + \beta_9 \text{Education}_i + \beta_{10} \text{VisMin}_i + \beta_{11} \text{VictoryMargin}_i + \beta_{12} \text{Density}_i \\
\log(\phi_i) &= \gamma_0 + \gamma_1 \log(n_i)
\end{aligned}
$$

where:

- p_i is the proportion of the total contributed amount that was sent out-of-district by donors in district i.
- μ_i and ϕ_i are the mean and common precision parameters of the distribution of p_i.
- n_i is the number of donations originating from district i.
- Amount_i is the total amount contributed by donors in district i.
- Region_{ji} are indicators for the donor region of district i.
- Age_i is the mean age of residents in district i.
- Income_i is the median household income in district i.
- Education_i is the proportion of residents in district i holding a post-secondary qualification.
- VisMin_i is the proportion of residents in district i who identify as a visible minority.
- VictoryMargin_i is the difference between the percentage of votes received by the winning and runner-up candidates, averaged across all three general elections in district i.
- Density_i is the population density (persons per km^2) of district i. As mentioned in-text, the model was fit using the square root of this variable.

4 Results

Here we present the results from our out-of-district propensity models. We note that many of our key findings are best understood as data summaries and visualizations, which are presented in Section 3.2. Section 3.2.1 provides context for how the localized contributions considered in our study fit into the broader scope of financial contributions in general. Section 3.2.2 provides spatio-temporal summaries of out-of-district propensity (which were used to inform the model discussed here). Section 3.2.3 briefly characterizes the funding network formed by out-of-district donations. Both summary-based and model-based findings are discussed together in Section 5.

Regression results from the individual- and district-level models are presented in Table 5 and Table 6 respectively. Figure 5 and Figure 6 provide a visualization of coefficient signs, magnitudes, and significances. For both models, exponentiating the coefficients yields a more interpretable value, the odds ratio. For the individual (i.e. logistic) regression model, an odds ratio above one for a categorical variable (e.g. recipient belonging to the Liberal Party) indicates that, holding all else constant, donations of that category have increased odds of being out-of-district, relative to the baseline category (e.g. recipient belonging to the Conservative party). For the district level (i.e. beta) regression model, an odds ratio above one for a categorical variable (e.g. district situated in the Prairies) indicates that, holding all else constant, the expected ratio of out- vs. in-district contributions increases. Odds ratios less than one and continuous variables (e.g. median household income) are interpreted accordingly.

Individual donations are less likely to reach out-of-district when they are bound for candidates instead of district associations (EDAs), and more likely for Liberal recipients instead of Conservative recipients. The contribution likelihoods of Conservative and N.D.P recipients were not meaningfully different. Larger donations were more likely to leave the district compared to smaller ones, helping to explain why amount-weighted proportions in our data summaries were consistently larger. The average margin of victory (taken to be a measure of district non-competitiveness) was, expectedly, positively associated with out-of-district propensity. Donations originating from all regions were less likely to reach out-of-district compared to Ontario, with the exception of Quebec, whose donors have nearly 1.5 times the odds to donate out-of-district, holding all else constant. Donations originating in the Territories were especially unlikely to travel outside their district, with nearly one-eighth the baseline Ontario odds. Turning to the donor's socio-demographic environment, regions with higher population densities, post-secondary education rates, median household incomes, and visible minority proportions[26] all favoured out-of-district donation. However, the average age did not significantly impact out-of-district chances in the presence of other socio-demographic variables[27]. When ethnicities were disaggregated, the positive effect of the visible minority proportion was observed for all minority ethnic groups recorded in the Census (see Appendix D).

At the district level, we modelled the ratio of out- to in-district contribution sums. Similarly to the individual models, districts with higher population densities, post-secondary education rates, median household incomes, and visible minority proportions were more likely to allocate more of their contributions to political entities in other districts rather than their own. The average age of a district was, similarly, insignificant in the presence of other socio-demographic variables. Quebec and the Territories were the only regions to differ significantly from Ontario in their out-of-district ratios, and with the same coefficient signs as in the individual model. Unlike the individual model, the total amount of donations raised and the average margin of victory in a district did not have a significant effect on the proportion of that amount that left said district; while donations are more likely to travel out-of-district when they are larger, and when districts are less competitive, these effects seems to be a purely individualized.

[26]This effect was observed even when donations from the four Brampton constituencies were removed, since data summaries showed that these clearly had very high influence on the model fit. A similar check was performed on the district-level model.

[27]Average age was significant in individual and district-level models where it, along with population density, were the only socio-demographic predictors. This tells us that its effect is mostly explained by other factors, such as income or educational attainment.

The eight variables used in our district-level model were able to explain a large amount of the variance in the ratio of out- to in-district donation ($pseudoR^2 = 0.67$). The same model fit on only population density also achieved a relatively good fit ($pseudoR^2 = 0.45$), however other variables, and especially socio-demographic ones such as educational attainment and visible minority proportion, improved the model's explanatory power well past what population density alone could offer.

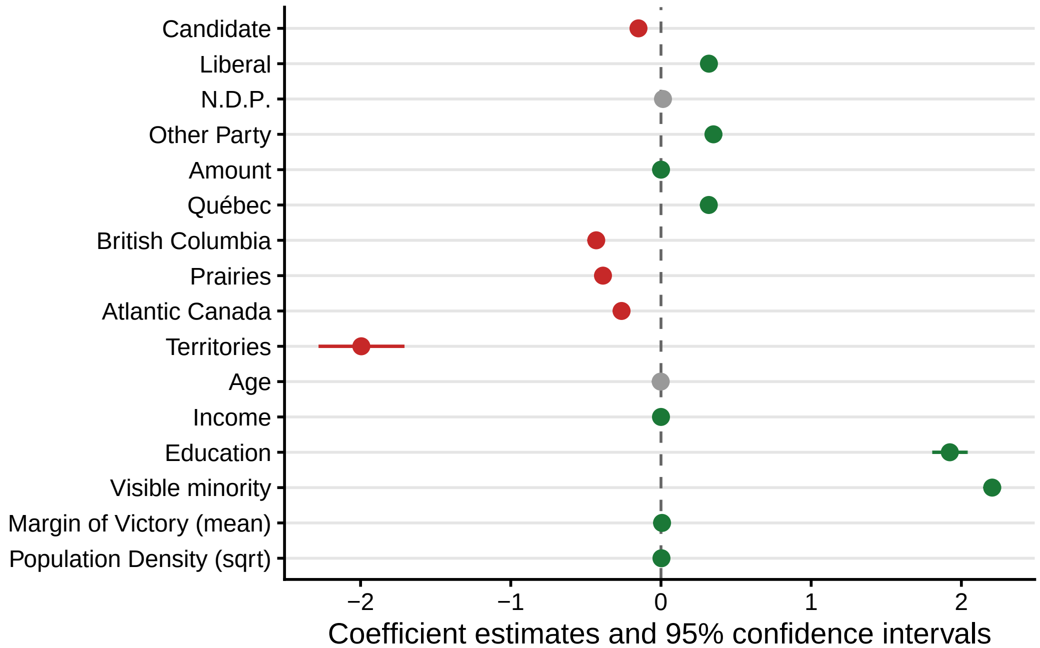


Figure 5: Coefficient estimates and 95% confidence intervals from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Coefficients are coloured grey if they are insignificant at alpha = 0.05, red if they are significant with a negative effect, and green if they are significant with a positive effect. Reference entity is the district association, reference party is the Conservative Party, reference region is Ontario.

5 Discussion

In this paper, we used publicly available political financing data to document the extent and spatio-temporal trends in Canadian out-of-district donations from 2015 to 2024. Further, we identify environmental characteristics associated with out-of-district propensity, quantify the reach of these donations, and provide an initial look at the spatial network formed by out-of-district donations. Our findings are also contextualized within the broader scope of all Canadian federal political contributions over $200 during this time (see Section 3.2.1). We present and discuss our major findings here, along with some notable study limitations.

During our study period, out-of-district donations made up a large share of all donations

Table 5: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. The reference entity is the district association, the reference party is the Conservative Party, and the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of entries) were incurred due to missingness in the census data.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−2.889**	**0.062**	**0.056**	**0.003**
District Associations	-		-	
Candidates	**−0.150**	**0.015**	**0.861**	**0.013**
Conservative	-		-	
Liberal	**0.319**	**0.014**	**1.376**	**0.020**
N.D.P	0.013	0.018	1.013	0.019
Other	**0.349**	**0.028**	**1.418**	**0.040**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.318**	**0.020**	**1.375**	**0.028**
British Columbia	**−0.431**	**0.018**	**0.650**	**0.011**
Prairies	**−0.386**	**0.019**	**0.680**	**0.013**
Atlantic Canada	**−0.262**	**0.027**	**0.769**	**0.021**
Territories	**−1.995**	**0.145**	**0.136**	**0.020**
Age	−0.002	0.001	0.998	0.001
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.923**	**0.059**	**6.841**	**0.406**
Visible Minority	**2.205**	**0.029**	**9.071**	**0.260**
Margin of Victory (mean)	**0.007**	**0.001**	**1.007**	**0.001**
Population Density (sqrt)	**0.003**	**0.000**	**1.003**	**0.000**
Num.Obs.	136 437		136 437	
AIC	164 482.3		164 482.3	
BIC	164 649.3		164 649.3	
Log.Lik.	−82 224.138		−82 224.138	
F	1115.752		1115.752	
RMSE	0.46		0.46	

Table 6: Results from a beta regression model where the outcome is the out-of-district share of the sum of contributions over $200 originating from an individual district during our study period. The reference entity is the district association, and the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Small losses (~ 1.1% of individual transactions) were incurred due to missingness in the census data. Each of the 338 districts was represented in this model.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−4.276**	**0.797**	**0.014**	**0.011**
Amount	0.000	0.000	1.000	0.000
Ontario	-		-	
Quebec	0.556	0.112	1.744	0.196
British Columbia	−0.166	0.098	0.847	0.083
Prairies	−0.144	0.109	0.866	0.094
Atlantic Canada	**0.228**	**0.138**	**1.256**	**0.173**
Territories	**−1.339**	**0.483**	**0.262**	**0.126**
Age	0.015	0.016	1.015	0.016
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.654**	**0.600**	**5.226**	**3.137**
Visible Minority	**1.918**	**0.221**	**6.805**	**1.505**
Margin of Victory (mean)	0.004	0.003	1.004	0.003
Population Density (sqrt)	**0.012**	**0.002**	**1.012**	**0.002**
Num.Obs.	338		338	
R2	0.672		0.672	
AIC	−473.8		−473.8	
BIC	−416.4		−416.4	
RMSE	0.13		0.13	

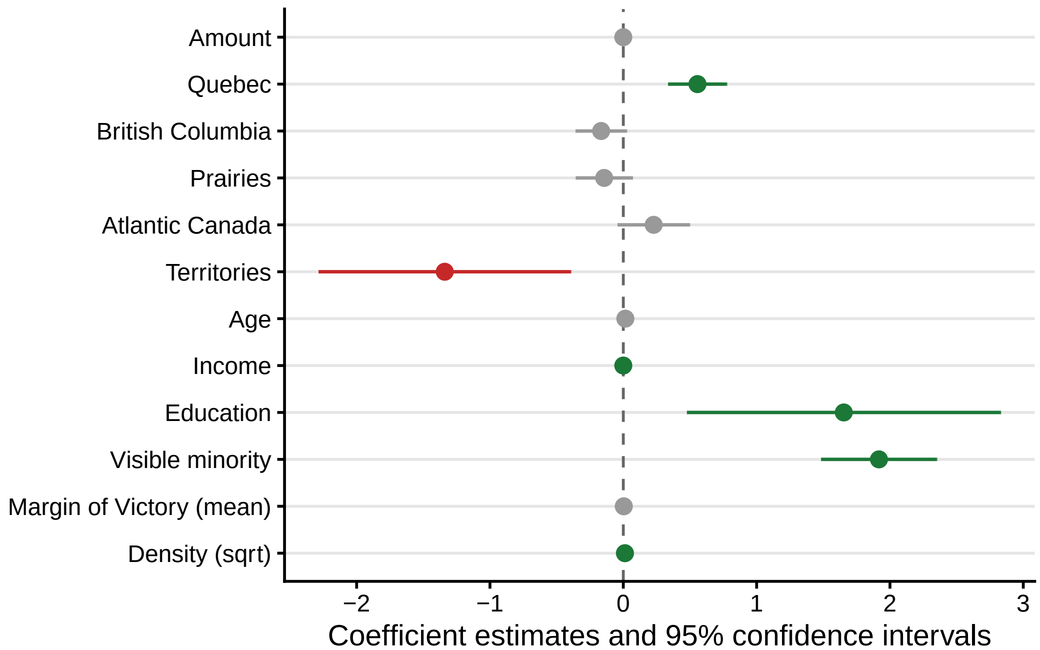


Figure 6: Coefficient estimates and 95% confidence intervals from a beta regression model where the outcome is the out-of-district share of the sum of contributions over \$200 originating from an individual district during our study period. Coefficients are coloured grey if they are insignificant at alpha = 0.05, red if they are significant with a negative effect, and green if they are significant with a positive effect. Reference entity is the district association, reference region is Ontario.

received by localized political entities (electoral candidates and district associations). Naturally, donation rates to these two political entities varied through time; campaign funding was restricted to election periods, and donations to district associations swelled during the holiday season and electoral cycles, bringing in nearly five times more contributions compared to baseline rates. The share of funding received from outside their district, however, remained relatively constant throughout the decade: roughly 40% of individual donations over $200, accounting for 44% of the total amount contributed. This rate and its consistency through time contrasts against U.S. findings, where the bulk of existing out-of-district donations research has been done. There, out-of-district funding shares to senators and congressional representatives during electoral campaigns have increased since the start of the 21st century (Gimpel et al. 2006; Canes-Wrone and Miller 2022; Sievert and Mathiasen 2023), and the average representative in 2016 received around two-thirds of their contributions from out-of-district (Gimpel et al. 2008; Canes-Wrone and Miller 2022).

While out-of-district contribution rates varied little over time, across parties, and between localized entities, we observed drastic differences in rates from district-to-district. For example, out-of-district donation shares to district associations over the entire studied decade were as little as 5% for some districts (e.g. Niagara Falls), and as high as 75% in others (e.g. Toronto — St. Paul's). While we might expect passively higher out-of-district rates solely due to the spatial proximity of other districts, empirical variation in rates of out-of-district giving (aggregated at the district level) far exceeded simulated expectations under this hypothesis. Further, most deviations from expected out-of-district donation rates were districts with higher than expected out-of-district contribution shares (see Appendix E). Also, this variation is not simply reflective of differences in the total amount each district gives; restricting our sample to only those districts among the top 10% by total contributed amount yielded a similarly broad range of out-of-district propensities.

Donations originating from more densely populated, higher-income, and better-educated districts are more likely to travel out-of-district, all else held constant. These findings agree with the profile of the "donor districts" identified in the U.S. (Gimpel et al. 2008), and with the predictors of donation propensity across North America in general (Gimpel et al. 2006, 2008; Garnett 2024). By contrast, the visible minority proportion of a donor's home environment is positively associated with out-of-district propensity, departing from the historically white profile of American "donor districts" (Gimpel et al. 2008). Recent Canadian studies support this inverted relationship between ethnicity and political engagement. Garnett et al. (2022), among others, note that some minority populations such as South Asians donate at rates which far exceed their share of the population, and, topically, Besco and Tolley (2022) found that individual donors belonging to most minority ethnic groups show higher rates of out-of-district donation compared to donors of European descent. The strong relationship between highly minoritized regions of the country and rates of out-of-district giving identified in our study is most plausibly attributed to one of two mechanisms which our models, as specified, cannot disentangle: demographic collinearity, and individual co-ethnic affinity. First, the relationship may simply reflect Canada's unique social demography. There, visible minority populations are concentrated almost exclusively in cities (Hou 2004), and higher minority composition is not obviously associated with lower income at the national level (Walks and Bourne 2006). In the U.S., however, visible minority proportion does not closely follow an urban-rural gradient (Lee and Sharp 2017), and these areas do have lower income, on average (Quillian 2012). Therefore, our visible minority covariate may be absorbing residual features of affluent, dense, urban Canada that density and income alone do not fully capture, rather than reflecting effects of ethnicity per se. Alternatively, this relationship may be signalling the presence of out-of-district co-ethnic donations. Identity-based donor affinity is a well-established feature of Canadian political participation (Besco 2019; Besco and Tolley 2022; Tolley et al. 2022). Further, the largest component of our inter-district donation network we could identify is the flow of funds between the four districts that make up the Brampton region within the Greater Toronto Area, whose population is majority South Asian. While attractive, this hypothesis cannot be confirmed without individualized ethnic identification data, which is largely inaccessible at this scale.

Despite empirically non-trivial rates of out-of-district donation, our study provides evidence that these donations remain a largely regionalized phenomenon. Nearly half of donations that leave their district during our study period were destined for entities in adjacent ones, and only around 12% ever even left their donor's province (in particular, 9% for candidate-bound donations). U.S. research on out-of-state donations describes highly nationalized donor bases among incumbent and campaigning House representatives (Gimpel et al. 2006; Canes-Wrone and Miller 2022; Sievert and Mathiasen 2023; Keena 2024). For example, a study on Senate elections from 2000 to 2020 found that both Republican and Democrat senators received, respectively, 60% and 49% of their donations from out-of-state during election periods (Sievert and Mathiasen 2023). The spatial proximity of U.S. states relative to Canadian provinces and territories may explain some of this difference, but its scale speaks to a more fundamental difference between the geographical donor bases of each country. The typical localized political entity in Canada sources donor-based funding largely from within their own province. Further research on this matter should examine whether more complex patterns exist in this donor network, such as the clear divide between sender and receiver districts displayed in Figure F8.

Our findings are accompanied by some important limitations. Following Gimpel et al. (2008) and Garnett (2024), donations under $200 (34% of the total amount sent to localized entities) were excluded from our study. The majority (75%) of exclusions were donations whose donors did not disclose their location. Comparisons between these contributions and those used in our study (see Appendix C) showed that they were similarly distributed across recipient characteristics such as party affiliation, region, and over time, suggesting that their exclusion introduced little systematic bias into our study dataset. However, 25% of donations under $200 were location-disclosed (8.5% of the total), and were distributed quite differently. To an overwhelming degree, these were in-district donations to Liberal district associations outside of election cycles. Further investigation revealed that only 12.4% of these transactions had a unique donor name, postal code and year, against 92.7% in our study data, suggesting that most come from donors whose annual total contributions exceeded the $200 disclosure threshold (Elections Canada 2026a). Their exclusion therefore likely inflates our out-of-district estimates for Liberal district associations, already the highest of any entity-party pair (57% against a 40% average). More generally, this finding, paired with the positive association between amount donated and out-of-district propensity, suggests that our estimates of out-of-district donation rates (particularly the amount-unweighted point estimates) are overestimating the true underlying out-of-district propensity. Nevertheless, since contributions under $200 remained similarly distributed to the study data, particularly in space, their exclusion is unlikely to have qualitatively affected the fitted coefficients of the most interesting covariates in our out-of-district propensity models, including those related to social demography.

More conceptually, studying out-of-district contributions meant restricting our focus to contributions sent directly from donors to localized political entities, precluding other essential pathways that govern campaign finance broadly. The most important of these excluded funding pathways are contributions to national parties and intra-party transfers. As shown in Section 3.2, most (~ 80%) contributions above $200 are received by national parties or leadership contestants, and roughly 50% of the time during general election periods. Because transfers limit the financial leverage of direct individual donations, the regionalization of donor bases we identify cannot be read as evidence that the overall funding of these entities is regionalized: in principle, localized entities may be routinely funded by party transfers originally sourced from donors thousands of kilometres away. Despite this, direct individual contributions are clearly an essential pathway for the financing of many local campaigns (Cross et al. 2020; Garnett 2024), especially when taking into account contributions to district associations (Cross et al. 2020), which we do. Given the prevalence of individual donations to national parties, and the essential part they play in local campaign financing, calls for further research investigating whether national party donor bases are mostly constrained to particular regions of the country, as has been found in U.S. research. Our accessible dataset of donor-geolocated contributions is well-positioned to answer this question.

Appendix

A Donations Data Pipeline

Here we describe the three major cleaning and validation steps carried out to transform contributions data from Elections Canada into the dataset used throughout our analysis. These scripts rely on other government datasets, cleaned separately, and documented in Appendix B.

A.1 Investigative Journalism Foundation Pre-Cleaning

The original contribution datasets used in this study can be found on Elections Canada's "Open data on contributions" page, under the heading "Data sets updated" (Elections Canada 2025a). Two products are offered. The first is a dataset of contribution reports filed by all Canadian registered federal political entities from January 2004 to present (Elections Canada 2025a). The second is an audited version of the first. Due to pre-screening policies and administrative lag, only some of the first dataset is audited. In our case, the audited version of the dataset had only 61.7% as many entries as the full un-audited data (n = 10095198; both accessed on May 5, 2026). Both datasets are updated on a weekly basis by Elections Canada (Elections Canada 2025a), as new audits and financial reports are filed.

Staff at the Investigative Journalism Foundation created a data pipeline to aggregate, de-duplicate, and clean these datasets. Both audited and un-audited versions of the same record were available for most records. To avoid duplicating these, audited and un-audited versions of the same record were linked by assuming that rows sharing the same donation date, electoral event, recipient, and recipient political party referred to the same entry. We note that this linkage approach carries the small risk that two distinct donations to the same recipient on the same date are incorrectly treated as duplicates. The pipeline preferentially keeps audited records when they are available.

Further de-duplication was effectuated differently based on the type of contribution being recorded. For loans, sums reported during an Annual filing might appear in a Quarterly report filed in the same year. In such cases, Annual records were preserved and Quarterly records were discarded. For all other types of contributions, only true duplicates (i.e. rows with identical entries in every field) were discarded. Additionally, some entries represented aggregated sums of contributions from anonymous donors. For these, Elections Canada reports a cumulative total per recipient Annually and Quarterly, meaning that the same aggregate sum could appear twice in the same year. To avoid double-counting these, only the largest reported aggregate sum per recipient per year was retained. Finally, a few smaller cleaning procedures were applied to the columns themselves; Table A1 shows where each of the columns present in the IJF data were drawn from columns at the source, and any notable procedures applied to them.

Table A1: Description of where the columns in the IJF data were derived from the source, and any cleaning procedures applied. Variables added for internal database consistency (e.g. processing dates) and redundant variables (e.g. donation year) are not shown here.

IJF Column	EC Source Column(s)	Notes
donor_full_name	Contributor name ; Contributor last name ; Contributor first name ; Contributor middle initial	Concatenated source columns, delimited by commas.
donor_type	Contributor type	No changes from source.
donor_location	Contributor City ; Contributor Province ; Contributor Postal code	Concatenated source columns, delimited by commas.
donation_date	Contribution Received date ; Fiscal/Election date	Used 'Contribution Received date' when available, filled with Fiscal/Election date otherwise. Flagged when imputation occurred. Removed invalid dates.
electoral_event	Electoral event	Donations to leadership and nomination contestants had NA entries here. Otherwise, no changes from source.
amount_monetary	Monetary amount	No changes from source.
amount_non_monetary	Non-Monetary amount	No changes from source.
recipient	Recipient ; Recipient last name ; Recipient first name ; Recipient middle initial	Concatenated source columns, delimited by commas.
electoral_district	Electoral District	No changes from source.
political_entity	Political Entity	No changes from source.
political_party	Political Party of Recipient	No changes from source.

A.2 Initial Cleaning of IJF Data

We applied further cleaning and filtering steps to the data supplied by the Investigative Journalism Foundation. First, we removed redundant columns (e.g. data had columns `year` and `donation_year` which were identical for every entry) and columns used for internal consistency (e.g. the date the entry was added by the organization). Next, we removed entries with malformatted dates, or disagreements between the year and date columns. We then filtered for only donations stated to have been made by individuals. Roughly 0.4% of the data was removed at this step, the overwhelming majority of which were records dated before 2006, outside the scope of our study. Next, we filtered for donations where the sum of monetary and non-monetary contributions was present and positive. We then removed entries where the donor or recipient name was either missing or shorter than 4 characters (after trimming whitespaces). Finally, we removed individual donations with amounts larger than $25000. Here we were careful to exclude entries that represented aggregated sums of anonymous contributions, which were flagged by searching for entries where `donor_full_name` started with "Contribut". In total, roughly 0.41% of entries (n ~ 30,000) were discarded in this section.

Donor postal codes were extracted from the `donor_location` column using regex-based string matching. Canadian postal codes always follow the format A0A0A0, where 0 is a digit and A is a letter (excluding I, O, D, F, Q, or U). Also, they may not start with W or Z (Canada Post 2026). These facts were used to verify the structure of extracted postal codes, and apply the following imputations:

- "O"s in a number slot were taken to mean "0"
- "I"s in a number slot were taken to mean "1"
- "!" in a number slot were taken to mean "1"

After imputations, we were able to recover 99.89% of individual donor postal codes. We note that while these postal codes are guaranteed to be structurally correct, we did not validate whether the codes matched their associated provinces, cities, or were actually used. The effective validity of these postal codes is verified in Appendix A.3. 0.35% of entries (n ~ 26,000) represented aggregated sums of anonymous contributions, and therefore did not have associated postal codes. Despite their scarcity, these amounted to 24.1% of the total contributed amount in the dataset. These were kept at this stage.

We also performed simple string cleaning procedures such as whitespace stripping and entity matching (e.g. United Party of Canada (UP) and United Party of Canada clearly refer to the same political party). Notably all donation recipients in the dataset had associated electoral districts (apart from leadership contestants and registered parties, since these are non-local entities) and political parties. Spot checking revealed no district or party attribution errors among a small set of recipients, however we could not completely confirm that recipient districts or parties were completely correct.

A.3 Clean IJF to `donations_data`

In this section we describe how our primary study dataset was derived from the cleaned IJF data, and augmented to include the donor's district using a custom postal code conversion file (PCCF) described in Appendix B.1. The resulting dataset represents all donor-geolocated contributions above $200 made out to federal political entities (excluding nomination contestants and candidates during by-elections).

First, we restricted the dataset to only those entries with donation dates between October 2, 2015, and April 22, 2024, capturing the effective period of the 2013 representational order. Next, we removed entries where the recipient's district was not one of the 338 possible districts in the 2013 representational order, referencing our modified PCCF. Next, we removed entries where the reported electoral event did not happen within our date range (e.g. 39th general election was in 2006). Although it is possible that these were retroactively added for past events, less than 0.001% of the data had this issue, so their removal did not impact our analysis. Next, we removed individual donations addressed to localized entities which had amounts larger than what the contribution limits allowed. For candidates this was $5000, and for registered associations this was $3550. Next, we removed donations addressed to leadership contestants for parties where no such contest occurred during our study period. Nomination contestants were removed entirely, to simplify analysis. 3% of donations (7% of summed total) received during general election periods had receipt dates outside the stated period (usually dated after the writ period had ended, not before it started). These were removed when fitting our regression models, but kept in most descriptive visualizations.

As mentioned in the main text, our study considers only donations above $200. This discards 45% of the total contributed amount during our study period, and 32% of the total amount received by localized entities (i.e. candidates and district associations). These excluded data are discussed further in Appendix C. For the remaining entries, donor postal codes were matched to their corresponding census dissemination areas (DAs) and federal electoral districts (FEDs) using our modified PCCF file. 1.2% of donor postal codes (representing 1.3% of the total amount contributed) were either missing or could not be matched, and were dropped (2.1% for localized entities). Figure A1 breaks down the discarded data by its cause, for localized entities. Finally, columns were renamed, merged, and variables were joined to improve the dataset's usability. The resulting schema is shown in part in Table A2.

B Supplementary Datasets

In this section we describe how we created two custom lookup tables that the cleaning scripts in Appendix A rely on; one to convert between different levels of spatial hierarchy, and a second to retrieve specific socio-demographic statistics at the census dissemination area.

Table A2: Primary variables used in our study, with examples and descriptions. Although not shown, additional information used such as census variables or geographic attributes of districts are easily retrieved by relating a donor's dissemination area or district to custom-made lookup tables.

Variable	**Example**	**Description**
donor_name	Alexandra Lulka	Donor full name.
donor_dissemination_area	35202387	Dissemination area (UID) associated with donor postal code.
donor_district	Eglinton–Lawrence	Electoral district associated with donor postal code.
donation_date	2015-10-18	Date donation was received.
electoral_event	General Election	Electoral event (if any) occurring during donation.
total_amount	563.3	Summed monetary and non-monetary value of the donation, in CAD.
recipient_name	Joe Oliver	Recipient full name.
recipient_district	Eglinton–Lawrence	District of the recipient (if applicable).
political_entity	Candidates	Entity type of the recipient.
political_party	Conservative Party of Canada	Recipient party affiliation (if any).
is_out_of_district	FALSE	Whether donor and recipient districts are different.

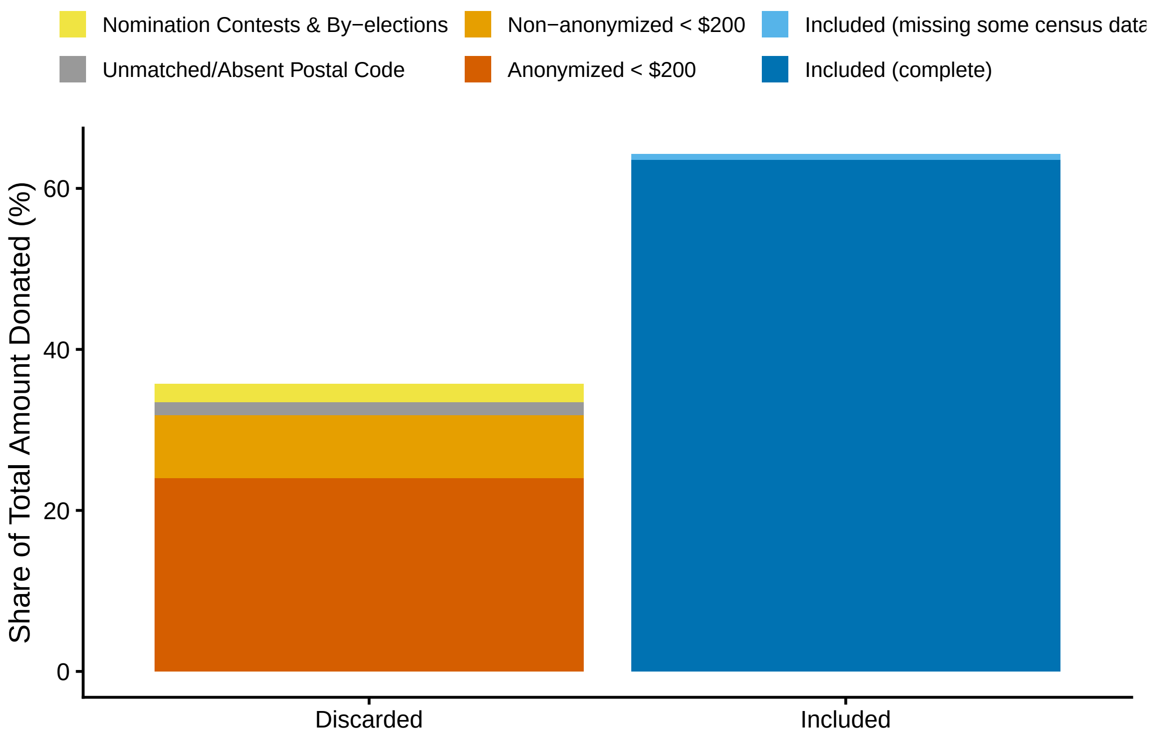


Figure A1: Total amount donated to localized entities (candidates and district associations) during our study period, split by whether or not the share was included in the study. The share of discarded contributions (left) is further broken down by the cause of its removal. We also display the share of included data to which not all census variables could be attributed (light blue).

B.1 Spatial Lookup Table

The purpose of the table constructed in this section is to provide unambiguous links between any administrative area of interest and another at a larger scale (e.g. postal code to district, dissemination area to province, etc.). This custom table is a version of Statistics Canada's 2024 postal code conversion files (PCCF), modified to resolve many-to-many spatial overlaps between postal codes, dissemination areas (DAs), and electoral districts (FEDs). The PCCF we used provides linkages between all registered Canadian postal codes (accurate as of June 2024) and various administrative divisions, including the dissemination areas (DAs) of the 2021 Census, and the federal electoral districts (FEDs) of the 2013 representational order (Statistics Canada 2025). These sets of DAs and FEDs are the same versions we chose for our study. Access to this PCCF was granted to us by the University of Toronto. The result of this process is shown in part in Table B1.

To resolve cases where one postal code straddled multiple dissemination areas, we used a single link indicator (SLI) to establish a one-to-one relationship between postal codes and DAs, available in PCCF files by default (Statistics Canada 2025). Statistics Canada determines SLIs by estimating the population that resides in each DA within the postal code area (based on 2021 Census population counts) and assigns the code to the one with a higher population (Statistics Canada 2025). This procedure removed 35% of links present in the original PCCF, however no single postal code was removed from the file entirely. A small number of postal codes (0.7%) were removed for missing either an associated DA or FED. All postal codes had associated provinces. In addition, 1.6% of dissemination areas in the dataset straddled multiple district boundaries. To resolve this, for each overlap case, we (1) counted how many postal codes within the dissemination area linked to each candidate FED, and (2) assigned the entire DA, and all of its postal codes to the FED holding the plurality of those postal codes, discarding the postal codes that mapped to any minority FED. Ties between FEDs were broken deterministically by FED UID. This resolution step resulted in the loss of roughly 0.5% of postal codes. Applied to the contributions dataset used in this study, these resolutions increased the share of entries whose donor postal codes could not be matched to the PCCF from 0.7% to 1.2%.

As a final step, we ran a suite of tests, primarily to confirm that the resolved table formed a strictly nested hierarchy: each postal code mapped to a single DA, each DA or postal code to a single FED, and each FED, DA, or postal code to a single province. We also verified that every postal code appeared exactly once, and that they were correctly formatted following Canada Post's guidelines (Canada Post 2026). We also confirmed that all DA and FED UIDs could be cross-referenced with Statistics Canada's 2021 DA and 2013 FED shapefiles (Statistics Canada 2022a), and that all 338 districts, 10 provinces, and 3 territories remained represented. Because of these tests, this spatial lookup table was treated as an authoritative list of valid postal codes, dissemination areas, districts, and provinces, which greatly simplified our analysis.

Table B1: Custom spatial lookup table, modified from Statistics Canada's Postal Code Conversion File. This table unambiguously links each Canadian postal code to a 2021 census dissemination area, and each census dissemination area to a federal electoral district from the 2013 representational order. Only a sample of four entries are shown.

Postal Code	Dissemination Area	Federal Electoral District
M4P1E4	35202387	Eglinton–Lawrence
K1A0A6	35061205	Ottawa Centre
V6B1A1	59153201	Vancouver Centre
H3A0G4	24661048	Outremont

B.2 Census Lookup Table

The purpose of the table constructed in this section is to provide specific socio-demographic variables (listed in Table B2) for every dissemination area (DA) present in the spatial lookup table (described in Section B.1). Socio-demographic information at this spatial scale was sourced from the 2021 Census, and specifically from the "Canada, provinces, territories, census divisions (CDs), census subdivisions (CSDs) and dissemination areas (DAs)" available for download online (Statistics Canada 2022b). We chose the 2021 Census over the 2016 Census since the midpoint of our time period (2019) is closer to 2021 than 2016. We chose the dissemination area, an administrative region that houses roughly 400 to 700 people, since it is the smallest spatial scale where each of our desired statistics are recorded (Statistics Canada 2021). All DAs listed in our spatial lookup file (see Appendix B.1) were present in the Census dataset we used. Since these DAs are the only ones we would ever use for analysis, the other 15.1% of DAs were dropped.

3.2% of remaining DAs were missing at least one census variable: 0.05% were missing all demographic variables, 0.43% only had population, area, and density, 0.12% were only missing the 25% sampled variables (i.e. ethnicity and education level), and 2.62% had all variables recorded except household income. Notably, all entries' missingness could be grouped into five strictly hierarchical stages, summarized in Table B3. Entries with missing variables were not dropped, but were instead flagged so that they could be easily identified during analysis. We also converted ethnicity and education counts to proportions. Table B4 provides a glimpse of the resulting Census dataset.

C Characterizing Excluded Contributions Under $200

In our analysis of out-of-district political donations addressed to electoral candidates and district associations, we excluded donations under $200, and donations whose donors could not successfully be geolocated, which in total represents 33.9% of the total amount received by

Table B2: Census IDs, descriptions, and sample sizes for each census variable used in this study. Variables with a 25% sample size were recorded for only 25% of the population, to preserve privacy.

Characteristic ID	Description	Sample Size
1	Population, 2021	100%
6	Population density (per square kilometres)	100%
7	Land area (square kilometres)	100%
39	Average (mean) age	100%
243	Median household income (total, in 2020)	100%
1683	Total population in private households	25%
1684	Total visible minority population	25%
1685	Total South Asian population	25%
1686	Total Chinese population	25%
1687	Total Black population	25%
1688	Total Filipino population	25%
1689	Total Arab population	25%
1690	Total Latin American population	25%
1691	Total Southeast Asian population	25%
1692	Total West Asian population	25%
1693	Total Korean population	25%
1694	Total Japanese population	25%
1998	Total population aged 15+	25%
2001	Total population with a postsecondary certificate	25%

Table B3: Structure of missing Census data. Missingness is hierarchical in the Census dataset: entries at a higher stage of missingness are always missing everything in the lower stages.

Missing Stage	Count	% of Total
0. Complete	47593	96.78%
1. Missing Median Household Income	1288	2.62%
2. Missing 25% Sample Variables	57	0.12%
3. Missing Age	211	0.43%
4. Missing Population	26	0.05%

Table B4: Custom census lookup table. This table provides desired socio-demographic information for every dissemination area in the 2021 Census. Only a sample of four entries are shown. Only population, density, and median household income are shown.

Dissemination Area (UID)	**Population**	**Density (per sqkm)**	**Median Household Income (CAD)**
35202387	612	4381.2	$98,500
35061205	548	3920.7	$87,200
59153201	701	6210.5	$76,300
24661048	489	5104.9	$92,100

Table C1: Breakdown of all contributions addressed to district associations and candidates (during general election periods) from 2015 to 2024, by cause of exclusion from analysis. Study Data is included for reference. We display the share of the total amount received by these entities capture by each case.

Data Case	Share of Total Contribution Amount
Study Data	66.1%
Unsuccessfully Donor-Geolocated (Over $200)	1.6%
Donor-Geolocated (Under $200)	7.9%
Anonymous (Under $200)	24.4%

these entities during our study period (see Table C1). In this section, we compare the recipient characteristics of contributions under $200 with those kept in our study to assess the impact of their removal on our analysis. By the *Canada Elections Act*, donors during a given calendar year (or writ period for donations to candidates) who contributed less than $200 in total are not required to have their name or address disclosed to Elections Canada (Elections Canada 2026d). Donations whose donors did not disclose this personal information are recorded in our dataset as aggregated sums, grouped by recipient and time-period (writ period for candidates, and quarterly or annual reports for district associations). Otherwise, donations under $200 appear as individual entries with associated donor names and locations. These two cases are compared against the data separately, and shown in Figure C1, and Figure C2 respectively.

Anonymous contributions under $200 are the largest source of missing data, and are overall similar in distribution to our study data across the factors considered (Figure C1). A larger proportion of these were addressed to district associations, Liberal entities over Conservative entities, and earlier in the decade (2015 to 2018, and 42nd general election) rather than later (2021 to 2023, and 44th general election). Anonymous contributions were nearly identically distributed to those used in our study with respect to the region of the receiving entity. No single factor among those considered in the anonymous contributions deviated from the study data by more than 10%.

Donor-geolocated contributions under $200 were distributed differently than the data in our study (Figure C2). A majority (83%) of these contributions were addressed to Liberal district associations specifically (see Panel E in Figure C2), largely outside general election periods. Further investigation into these cases revealed that only 12.4% of these transactions had a unique donor name, postal code and year, against 92.7% in our analysis data, suggesting that most come from donors whose annual total contributions exceeded the $200 disclosure threshold.

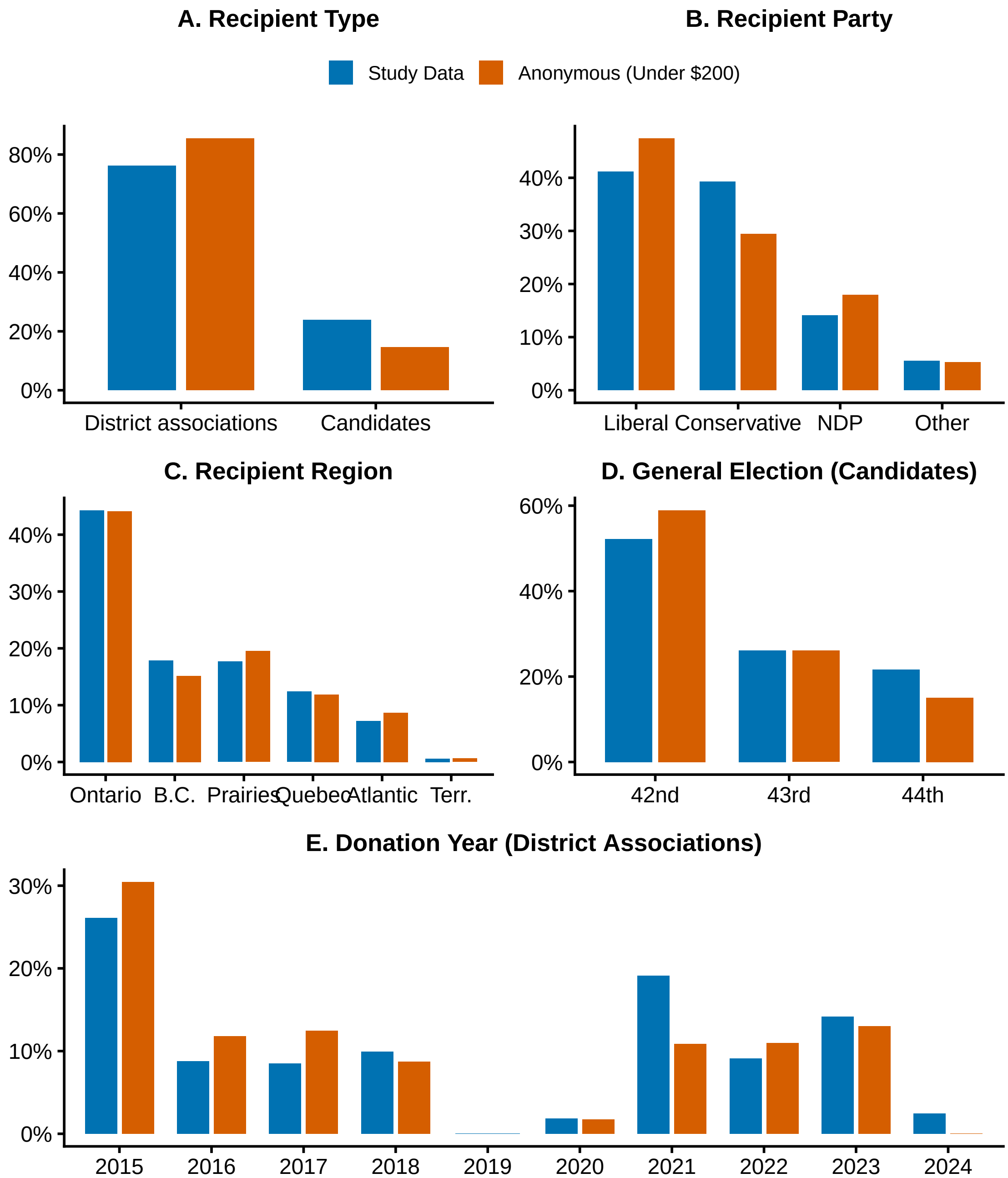


Figure C1: Comparing the distributions of donations to candidates and district associations above $200 whose donors were successfully situated in their districts (blue; those included in our study) vs. anonymous donations under $200 (orange; not included in the study). Datasets are compared across five factors, and distributions are weighted by total amount contributed. Panels D and E only compare datasets for donations bound for candidates and district associations respectively.

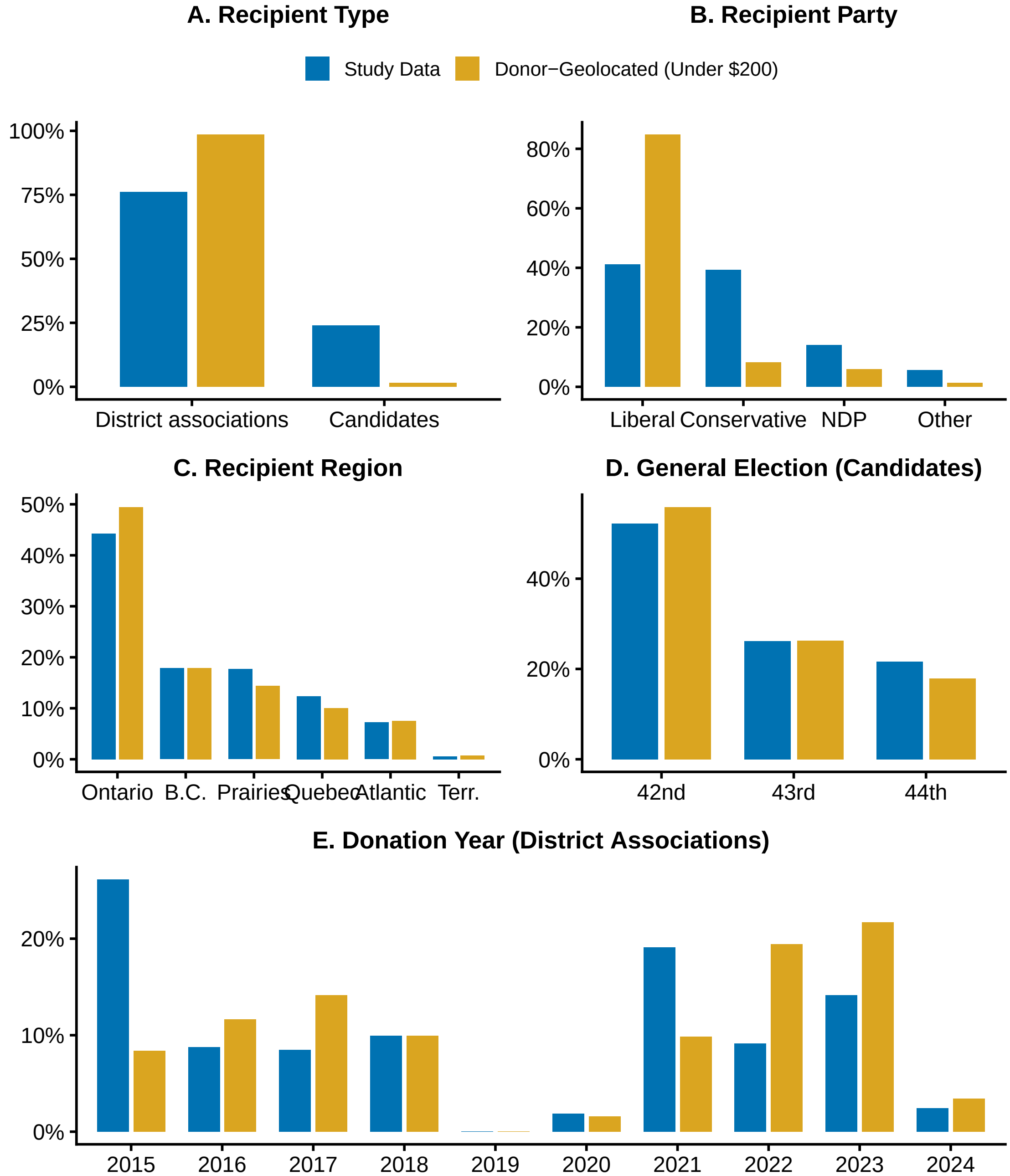


Figure C2: Comparing the distributions of donations to candidates and district associations above $200 whose donors were successfully situated in their districts (blue; those included in our study) vs. location-disclosed donations under $200 (yellow; not included in the study). Datasets are compared across five factors, and distributions are weighted by total amount contributed. Panels D and E only compare datasets for donations bound for candidates and district associations respectively.

D Alternate Model Specifications

Here we present the results of six alternate out-of-district propensity models, mainly fit to assess the robustness of our results in Section 4 to other, reasonable model specifications.

Table D1 shows results from an individual-level model with the same covariates, fit using only donations bound for district associations (i.e. excluding those received by candidates). We observe similar coefficients and overall model fit. The only major difference between this and the primary model is that average age is negatively associated with out-of-district propensity, with statistical significance at the $\alpha = 0.05$ level. Table D2 shows results from a district-level model with the same covariates, fit using only donations bound for district associations (i.e. excluding those received by candidates). We observe similar coefficients and overall model fit.

Table D3 shows results from an individual-level model with the same covariates, fit using only donations received by candidates (i.e. excluding those received by district associations). We observe similar coefficient and overall model fit. Table D4 shows results from a slightly more elaborate version of this model, that includes the election period in which the donation was received, as well as how competitive the donor's district was during that period. Both of these more specific covariates were significantly associated with out-of-district propensity, but did not affect the rough magnitudes and significances anywhere else in the model.

Table D5 and Table D6 shows results from individual level and district level models respectively, where province and visible minority proportion were disaggregated into more specific categories. In both models, the coefficients associated with provinces each behaved similarly to their parent region. The proportion of habitants belonging to all ethnic groups available in the Census were positively associated with out-of-district donation at the individual level. However, only the proportion of South Asian, Black, and Chinese inhabitants of a district was associated with it's share of donations that were bound for other districts. Further, disaggregation greatly increased the number of coefficients used to fit the model, for marginal gains in the model's explanatory power. For example, the disaggregated district level model has over double the number of coefficients compared to the one included in the main text, for only a single percentage increase in the model's explainability. For this reason, grouped variables were kept in the main models at both levels.

E Testing Whether Out-of-district Rates can be Explained by Distance Alone

E.1 Simulation Setup

Figure 3 shows how variable out-of-district donation rates are at the district level. However, this may simply be a byproduct of the spatial arrangement of electoral districts. Districts vary enormously in size, from the 13 square kilometres that make up Toronto–St. Paul's, to the

Table D1: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Only donations sent to district associations are considered. The reference party is the Conservative Party, and the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−2.789**	**0.069**	**0.062**	**0.004**
Conservative	-		-	
Liberal	**0.367**	**0.016**	**1.443**	**0.023**
N.D.P	0.020	0.022	1.021	0.022
Other	**0.153**	**0.038**	**1.166**	**0.044**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.294**	**0.023**	**1.342**	**0.030**
British Columbia	**−0.427**	**0.019**	**0.653**	**0.013**
Prairies	**−0.419**	**0.021**	**0.658**	**0.014**
Atlantic Canada	**−0.254**	**0.032**	**0.776**	**0.025**
Territories	**−1.842**	**0.152**	**0.159**	**0.024**
Age	**−0.004**	**0.001**	**0.996**	**0.001**
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.890**	**0.066**	**6.617**	**0.438**
Visible Minority	**2.054**	**0.032**	**7.801**	**0.252**
Margin of Victory (mean)	**0.008**	**0.001**	**1.008**	**0.001**
Population Density (sqrt)	**0.003**	**0.000**	**1.003**	**0.000**
Num.Obs.	107 442		107 442	
AIC	130 829.1		130 829.1	
BIC	130 982.5		130 982.5	
Log.Lik.	−65 398.556		−65 398.556	
F	914.461		914.461	
RMSE	0.46		0.46	

Table D2: Results from a beta regression model where the outcome is the out-of-district share of the sum of EDA-bound contributions over $200 originating from an individual district during our study period. The reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Each of the 338 districts is represented in this model.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−3.939**	**0.862**	**0.019**	**0.017**
Amount	0.000	0.000	1.000	0.000
Ontario	-		-	
Quebec	**0.538**	**0.120**	**1.713**	**0.205**
British Columbia	−0.182	0.104	0.834	0.087
Prairies	−0.205	0.117	0.815	0.095
Atlantic Canada	**0.301**	**0.151**	**1.351**	**0.203**
Territories	**−1.222**	**0.497**	**0.295**	**0.147**
Age	0.012	0.017	1.012	0.017
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.658**	**0.646**	**5.247**	**3.391**
Visible Minority	**1.847**	**0.238**	**6.340**	**1.511**
Margin of Victory (mean)	0.004	0.003	1.004	0.003
Population Density (sqrt)	**0.012**	**0.002**	**1.012**	**0.002**
Num.Obs.	338		338	
R2	0.607		0.607	
AIC	−401.2		−401.2	
BIC	−343.8		−343.8	
RMSE	0.14		0.14	

Table D3: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Only donations sent to electoral candidates during election periods are considered. The reference party is the Conservative Party, the reference region is Ontario. Estimates are bolded when significant at the alpha = 0.05 level.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−3.450**	**0.143**	**0.032**	**0.005**
Conservative	-		-	
Liberal	**0.079**	**0.037**	**1.082**	**0.040**
N.D.P	0.007	0.036	1.007	0.036
Other	**0.618**	**0.045**	**1.856**	**0.083**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.398**	**0.048**	**1.489**	**0.071**
British Columbia	**−0.438**	**0.043**	**0.645**	**0.028**
Prairies	**−0.292**	**0.044**	**0.747**	**0.033**
Atlantic Canada	**−0.237**	**0.051**	**0.789**	**0.040**
Territories	**−3.012**	**0.507**	**0.049**	**0.025**
Age	0.005	0.002	1.005	0.002
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**2.011**	**0.136**	**7.471**	**1.019**
Visible Minority	**2.636**	**0.063**	**13.960**	**0.878**
Margin of Victory (mean)	**0.005**	**0.001**	**1.005**	**0.001**
Population Density (sqrt)	**0.004**	**0.001**	**1.004**	**0.001**
Num.Obs.	28 995		28 995	
AIC	33 320.5		33 320.5	
BIC	33 452.9		33 452.9	
Log.Lik.	−16 644.236		−16 644.236	
F	273.999		273.999	
RMSE	0.44		0.44	

Table D4: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. Only donations sent to electoral candidates during election periods are considered. The reference party is the Conservative Party, the reference region is Ontario, and the reference general election period is the 42nd. Estimates are bolded when significant at the alpha = 0.05 level.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−3.339**	**0.143**	**0.035**	**0.005**
Conservative	-		-	
Liberal	0.059	0.037	1.061	0.039
N.D.P	−0.014	0.036	0.986	0.036
Other	**0.638**	**0.045**	**1.893**	**0.085**
42nd general election	-		-	
43rd general election	**−0.158**	**0.033**	**0.854**	**0.028**
44th general election	**−0.119**	**0.036**	**0.888**	**0.032**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Ontario	-		-	
Quebec	**0.402**	**0.047**	**1.495**	**0.071**
British Columbia	**−0.449**	**0.043**	**0.638**	**0.028**
Prairies	**−0.260**	**0.042**	**0.771**	**0.032**
Atlantic Canada	**−0.224**	**0.052**	**0.799**	**0.041**
Territories	**−3.028**	**0.507**	**0.048**	**0.025**
Age	0.005	0.002	1.005	0.002
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.989**	**0.136**	**7.307**	**0.995**
Visible Minority	**2.636**	**0.063**	**13.950**	**0.879**
Margin of Victory (per-cycle)	**0.003**	**0.001**	**1.003**	**0.001**
Population Density (sqrt)	**0.004**	**0.001**	**1.004**	**0.001**
Num.Obs.	28 995		28 995	
AIC	33 304.4		33 304.4	
BIC	33 453.3		33 453.3	
Log.Lik.	−16 634.198		−16 634.198	
F	242.454		242.454	
RMSE	0.44		0.44	

Table D5: Results from a logistic regression model where the outcome is whether or not an individual donation was sent out-of-district. The reference entity is district associations, the reference party is the Conservative Party, the reference province is Ontario. Estimates are bolded when significant at the alpha = 0.05 level.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−2.851**	**0.065**	**0.058**	**0.004**
Liberal	**0.328**	**0.014**	**1.388**	**0.020**
N.D.P	0.012	0.019	1.012	0.019
Other	**0.344**	**0.028**	**1.410**	**0.040**
Candidate	**−0.141**	**0.015**	**0.868**	**0.013**
Amount	**0.000**	**0.000**	**1.000**	**0.000**
Alberta	**−0.472**	**0.023**	**0.624**	**0.014**
British Columbia	**−0.429**	**0.019**	**0.651**	**0.012**
Manitoba	**−0.305**	**0.033**	**0.737**	**0.024**
New Brunswick	**−0.409**	**0.048**	**0.664**	**0.032**
Newfoundland and Labrador	**0.251**	**0.050**	**1.286**	**0.065**
Northwest Territories	**−1.665**	**0.259**	**0.189**	**0.049**
Nova Scotia	**−0.427**	**0.041**	**0.652**	**0.027**
Nunavut	**−2.306**	**0.423**	**0.100**	**0.042**
Prince Edward Island	**−0.373**	**0.070**	**0.689**	**0.048**
Quebec	**0.251**	**0.021**	**1.285**	**0.027**
Saskatchewan	**−0.277**	**0.038**	**0.758**	**0.029**
Yukon	**−2.110**	**0.193**	**0.121**	**0.023**
Age	−0.001	0.001	0.999	0.001
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.759**	**0.064**	**5.807**	**0.369**
South Asian	**1.988**	**0.045**	**7.304**	**0.325**
Chinese	**2.411**	**0.056**	**11.144**	**0.627**
Black	**2.533**	**0.147**	**12.596**	**1.851**
Filipino	**1.627**	**0.154**	**5.090**	**0.781**
Arab	**2.192**	**0.194**	**8.952**	**1.737**
Latin American	**6.726**	**0.330**	**833.532**	**275.452**
Southeast Asian	**3.916**	**0.341**	**50.222**	**17.134**
West Asian	**1.421**	**0.226**	**4.140**	**0.936**
Korean	**2.514**	**0.393**	**12.354**	**4.860**
Japanese	**3.253**	**0.824**	**25.859**	**21.304**
Margin of Victory (mean)	**0.008**	**0.001**	**1.008**	**0.001**
Population Density (sqrt)	**0.003**	**0.000**	**1.003**	**0.000**
Num.Obs.	136 437		136 437	
F	562.004		562.004	
RMSE	0.46		0.46	

Table D6: Results from a beta regression model where the outcome is the out-of-district share of the sum of contributions over $200 originating from an individual district during our study period. The reference province is Ontario. Estimates are bolded when significant at the alpha = 0.05 level. Each of the 338 districts is represented in this model.

	Coefficient		Odds Ratio	
	Est.	**S.E.**	**Est.**	**S.E.**
(Intercept)	**−4.926**	**0.875**	**0.007**	**0.006**
Amount	0.000	0.000	1.000	0.000
Alberta	−0.238	0.134	0.788	0.105
British Columbia	−0.061	0.140	0.941	0.132
Manitoba	0.141	0.180	1.152	0.207
New Brunswick	0.307	0.210	1.360	0.286
Newfoundland and Labrador	**0.518**	**0.240**	**1.679**	**0.403**
Northwest Territories	−1.134	0.774	0.322	0.249
Nova Scotia	0.157	0.199	1.170	0.232
Nunavut	−0.707	0.972	0.493	0.479
Prince Edward Island	−0.038	0.318	0.963	0.306
Quebec	**0.483**	**0.118**	**1.622**	**0.191**
Saskatchewan	0.171	0.189	1.187	0.224
Yukon	**−1.633**	**0.775**	**0.195**	**0.151**
Age	0.023	0.017	1.023	0.017
Income	**0.000**	**0.000**	**1.000**	**0.000**
Education	**1.977**	**0.722**	**7.218**	**5.212**
South Asian	**1.562**	**0.411**	**4.771**	**1.961**
Chinese	**2.250**	**0.444**	**9.484**	**4.215**
Black	**4.204**	**1.206**	**66.976**	**80.740**
Filipino	0.653	1.191	1.921	2.287
Arab	2.687	1.567	14.687	23.011
Latin American	3.856	4.289	47.256	202.694
Southeast Asian	6.267	4.493	527.095	2368.115
West Asian	0.402	2.303	1.494	3.442
Korean	4.480	4.205	88.201	370.871
Japanese	−6.419	17.602	0.002	0.029
Margin of Victory (mean)	0.004	0.003	1.004	0.003
Population Density (sqrt)	**0.010**	**0.003**	**1.010**	**0.003**
Num.Obs.	338		338	
R2	0.683		0.683	
RMSE	0.12		0.12	

entire territory of Nunavut. Assuming that social influence decays as a function of distance, we should expect a donor residing in a district in downtown Toronto to (1) be more aware of or (2) perceive to have a greater stake in district associations and candidates outside their own district, simply by virtue of their spatial proximity. By contrast, a donor in Nunavut may be expected to place less importance on political entities in the neighbouring Northwest Territories, whose population centre is nearly one thousand kilometres away. In this section we design a simulation to test the null hypothesis that the observed variance in district-level out-of-district donation rates is well-explained by differences in spatial clustering.

- Assign a random distance to each donation.
- If this distance is < 1km, immediately declare the donation as having been received in-district. Otherwise, continue.
- Assign a random direction for that donation.
- Determine if it landed within Canada. If it didn't choose a different direction. Otherwise, continue.
- Determine which district it landed in.
- Declare in-district vs. out-of-district.

Donation distances were assigned by sampling from the true distance distribution of the observed donations dataset, shown in Figure 4. The distance a donation travelled was set to zero if the donation was in-district, and otherwise set to be the spatial distance between the centroid of the donor's dissemination area, and the centroid of the recipient's district. *Statistics Canada* Shapefiles were used to determine which district a randomly travelling donation landed in.

For each district, 30 replicate simulations were run to estimate the sample distribution of the simulation. Using these distributions, the observed out-of-district proportions were compared with the simulated out-of-district proportions, for each district. Only results for the count of out-of-district donors (not the amount of donations sent out-of-district) was considered, however results were very similar for amounts, just with systematically wider sampling distributions.

E.2 Simulation Results

When considering donations addressed to candidates (aggregated over all three election periods), only 6.5% of districts fell within the expected proportion of out-of-district donations. For donations addressed to EDAs over the entire study period, only 5.3% of districts fell within the expected proportion of out-of-district donations. Proportions of districts that fell within their expected ranges were slightly higher (8.2% and 8.5% respectively) when looking at the share of the total contributed amount, however this was likely due to the systematically wider sampling distributions caused by the added variance incurred when using amounts.

Figure E1 shows that, for both entities, most districts have out-of-district proportions that deviate drastically from simulated expectations under the null hypothesis. Further, most

Table F1: Summaries of financial contributions to federal political entities over $200 from 2015 to 2024, separated by entity. Only donations stated to have been received during a general election period are shown. Contributions received by nomination contestants are not shown.

Political Entity	Type	Donation Count	Donation Amount (Thousands)	Share of Total Amount
Candidate	Localized	29,380	$18,872	25.7%
District Association	Localized	36,489	$20,373	27.7%
Leadership Contestant	National	134	$103	0.1%
Party	National	79,052	$34,071	46.4%

deviations, including the largest ones in magnitude, were districts where the observed out-of-district donation rate was larger than expected, as opposed to smaller. This indicates that most districts have higher out-of-district donation rates than what can be explained by the spatial arrangement of districts alone. This can be seen more clearly in Figure E2; in both cases, much of the support for the observed distribution lies strictly above the support for the expected distribution. This skew in deviations is especially noticeable for donations addressed to district associations. Results using amount-weighted proportions were similar in both cases.

F Additional Data Summaries

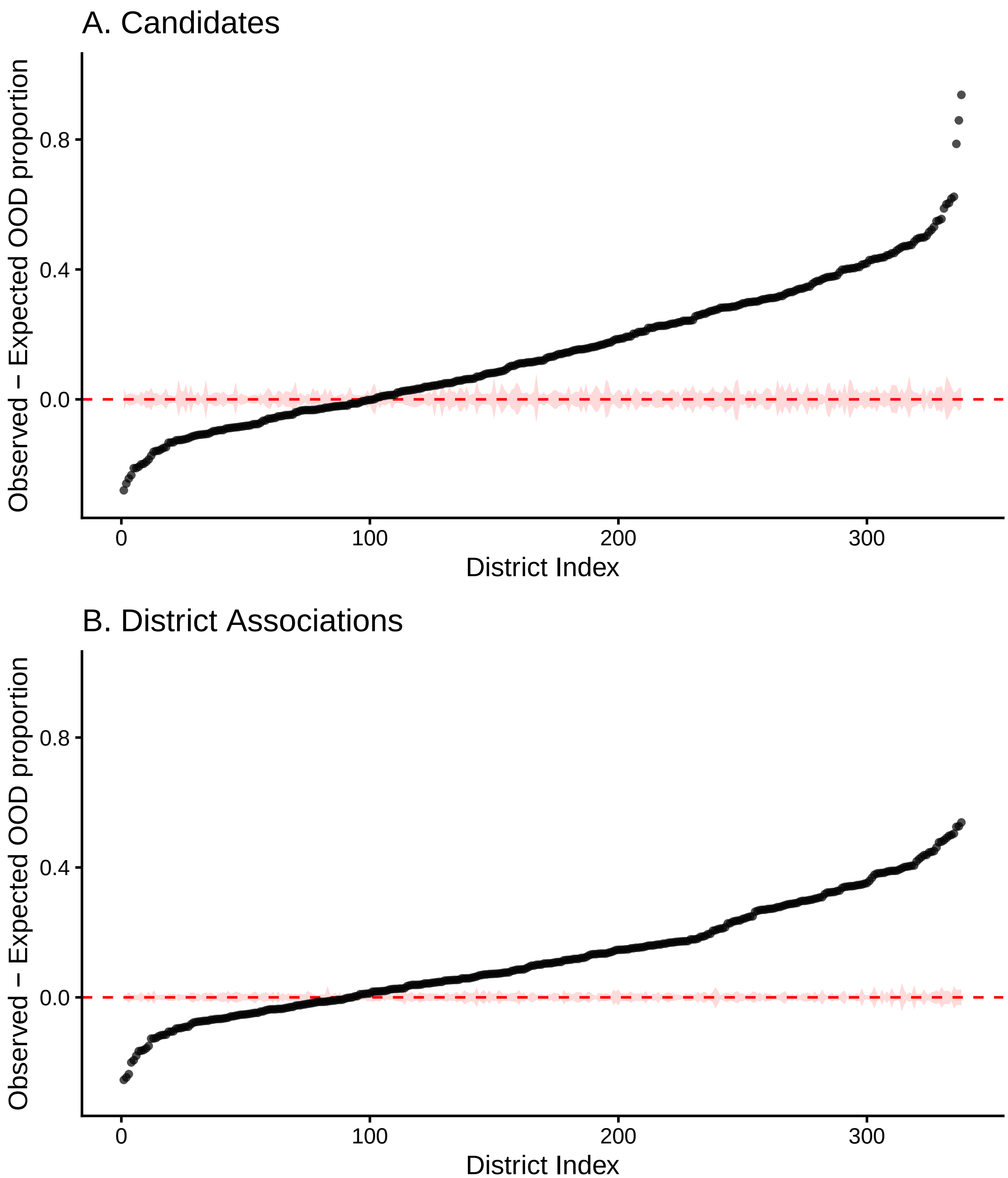


Figure E1: Deviation of each district's observed proportion of out-of-district (OOD) donations from its simulated proportion under the null hypothesis. Results for both donations to candidates (A) and district associations (B) are shown. Districts are sorted by the sign and magnitude of their deviation. The dashed red line indicates no deviation, and the shaded red band gives the simulated 95% confidence interval. Points outside the band deviate more than spatial clustering alone can explain.

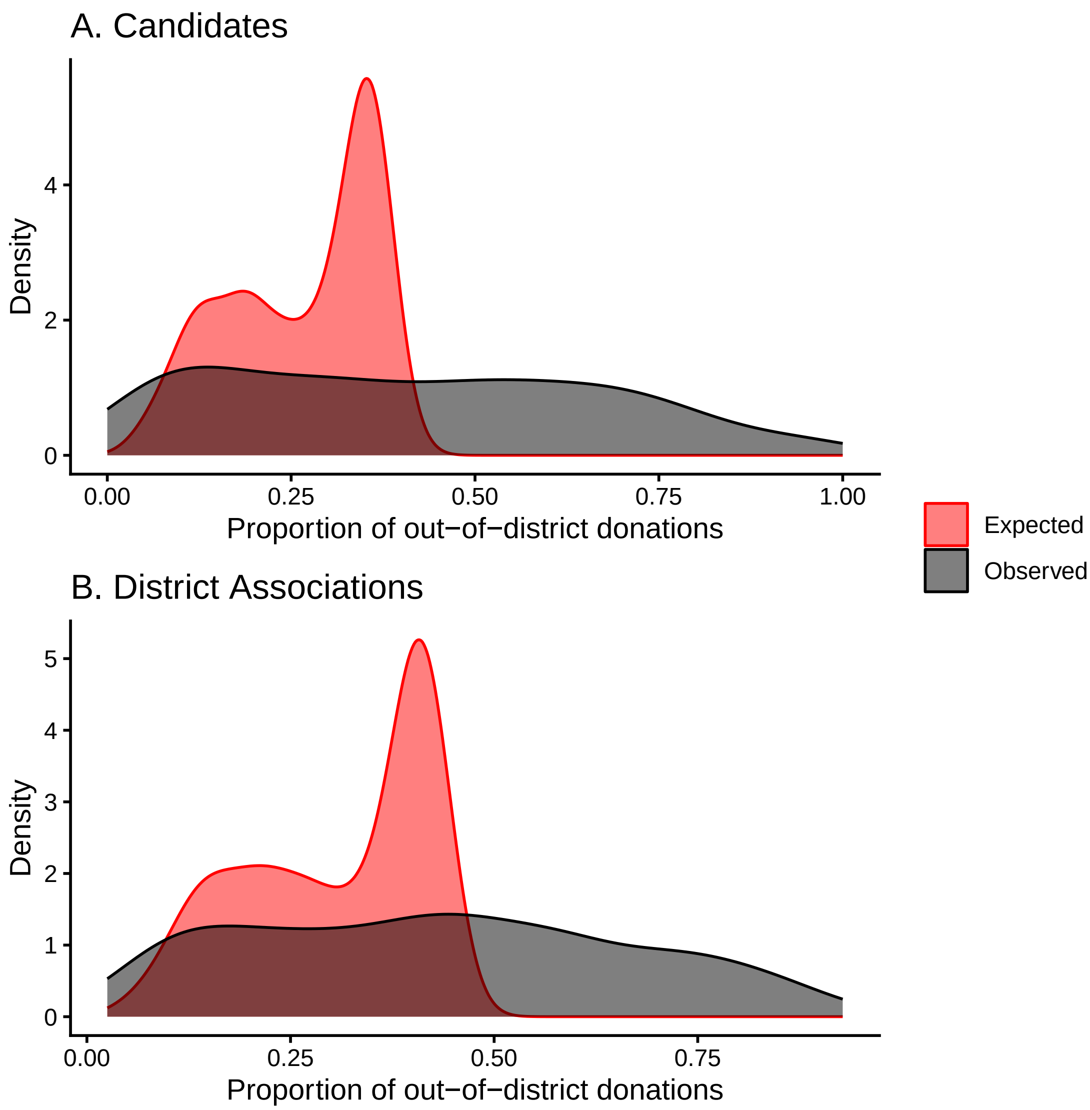


Figure E2: Distribution across districts of the observed (black) and simulated null-expected (red) proportion of out-of-district donations. Both donations to candidates (A) and district associations (B) are shown.

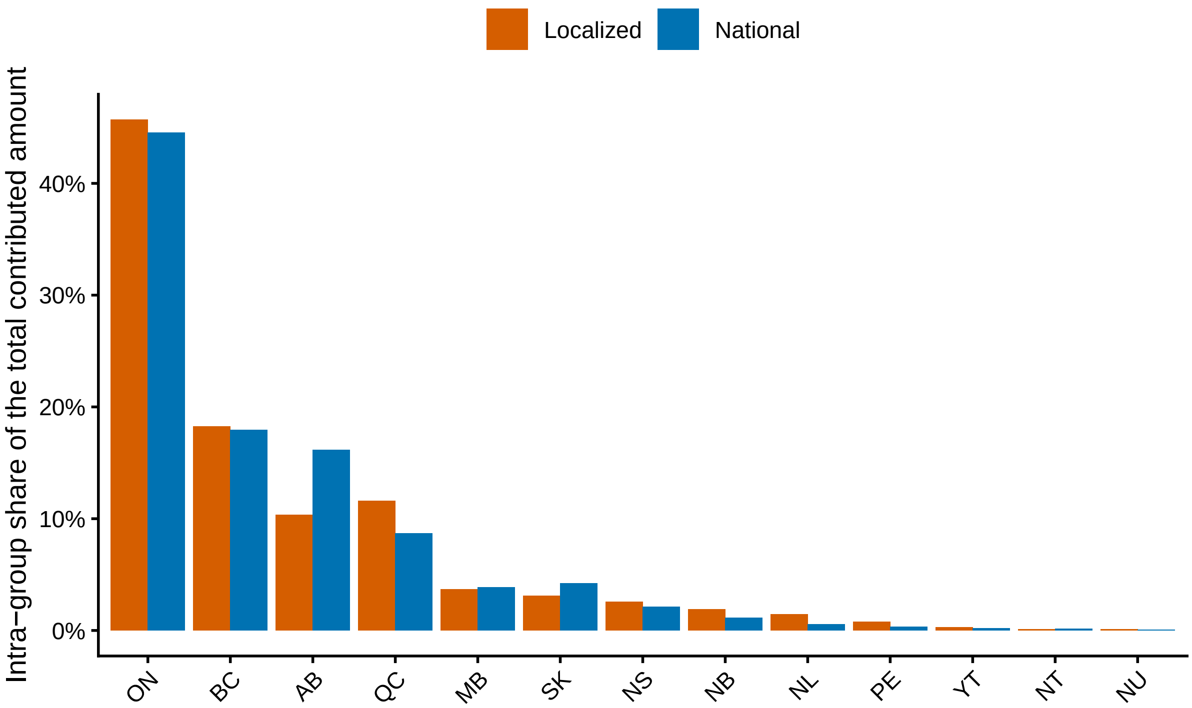


Figure F1: Distribution of contributions across donor provinces, shown separately for localized entities (candidates and district associations) and national ones (parties and leadership contestants). Within each group, bars give each province's share of that group's total contributed amount, and sum to 100% across provinces. Contributions over $200 received from 2015 to 2024.

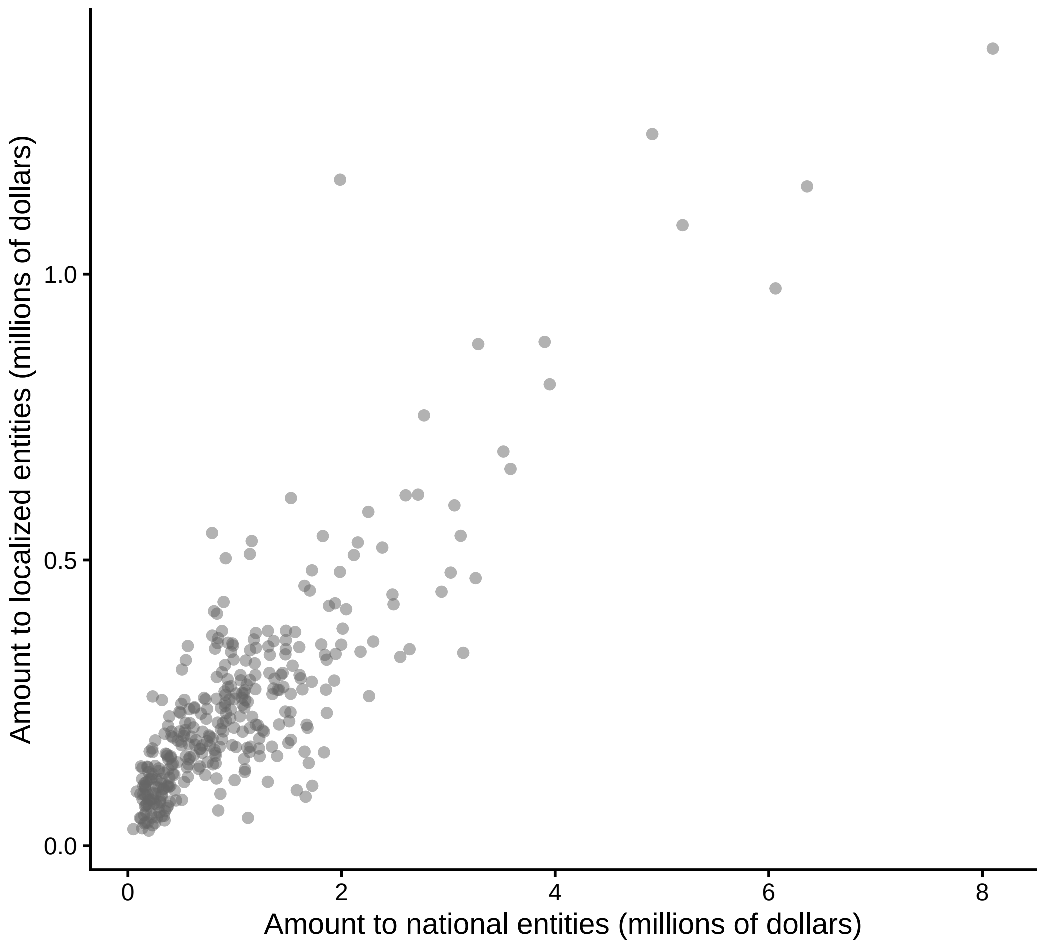


Figure F2: Total amount contributed to localized entities (candidates and district associations) against the total contributed to national entities (parties and leadership contestants), with one point per donor district. Contributions over $200 received from 2015 to 2024.

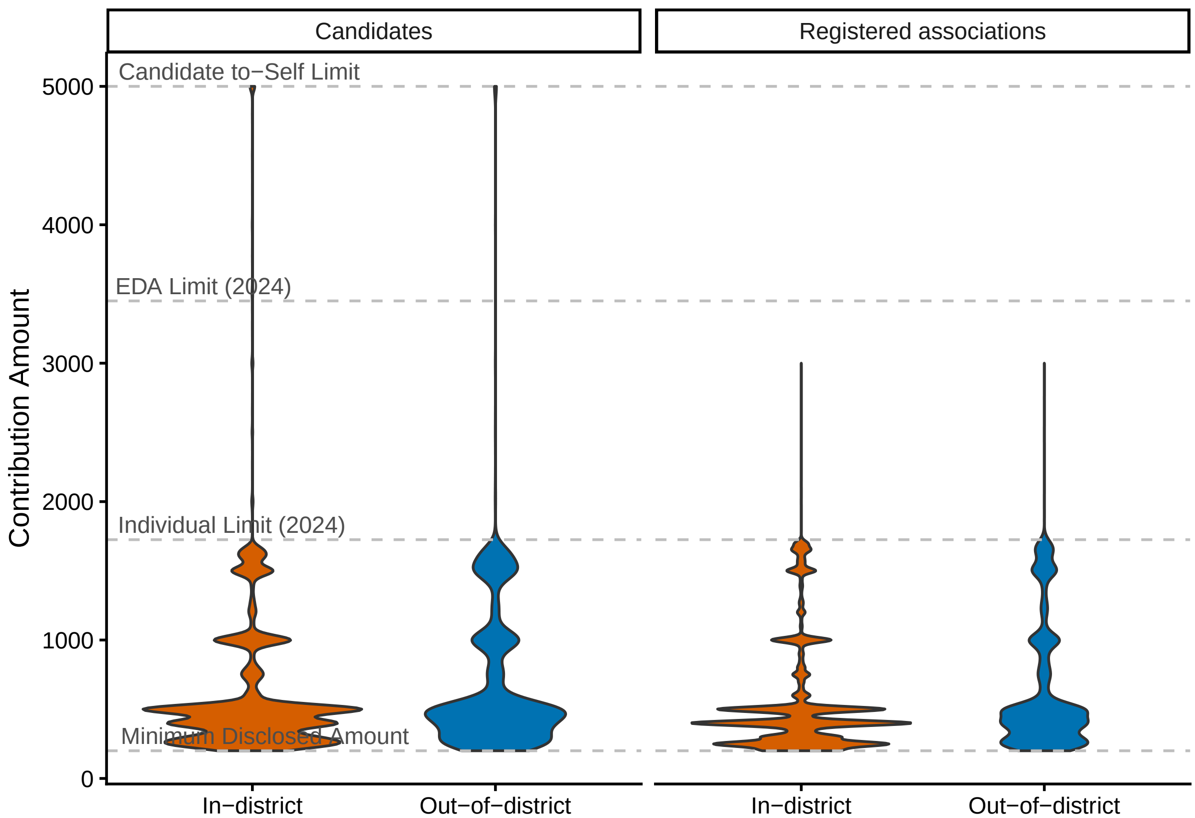


Figure F3: Value distributions of in- (blue) and out-of-district (red) contributions over $200 received by candidates (left) and district associations (right) from 2015 to 2024. Dashed grey lines mark notable individual contribution limits.

Table F2: Proportion of contributions over $200 that were received by localized entities (candidates and district associations) outside the donor's district, from 2015 to 2024, broken down by recipient entity and party. The three largest parties are shown individually; all others are grouped into 'Other'. Raw and value-weighted proportions of out-of-district donations (OOD) are reported, alongside overall contribution counts for reference. Contributions received by candidates outside general election cycles are not included in calculations.

Entity	**Party**	**Donation Count**	**OOD Proportion**	**OOD Proportion (Amount Weighted)**
Candidate	Conservative Party of Canada	14,421	36.1%	38.5%
Candidate	Liberal Party of Canada	5,961	39.7%	43.9%
Candidate	New Democratic Party	6,637	35.9%	39.6%
Candidate	Other	3,540	43.3%	41.4%
District Association	Conservative Party of Canada	38,756	36.0%	40.1%
District Association	Liberal Party of Canada	49,859	53.5%	57.8%
District Association	New Democratic Party	16,164	35.2%	35.8%
District Association	Other	3,840	34.6%	36.0%

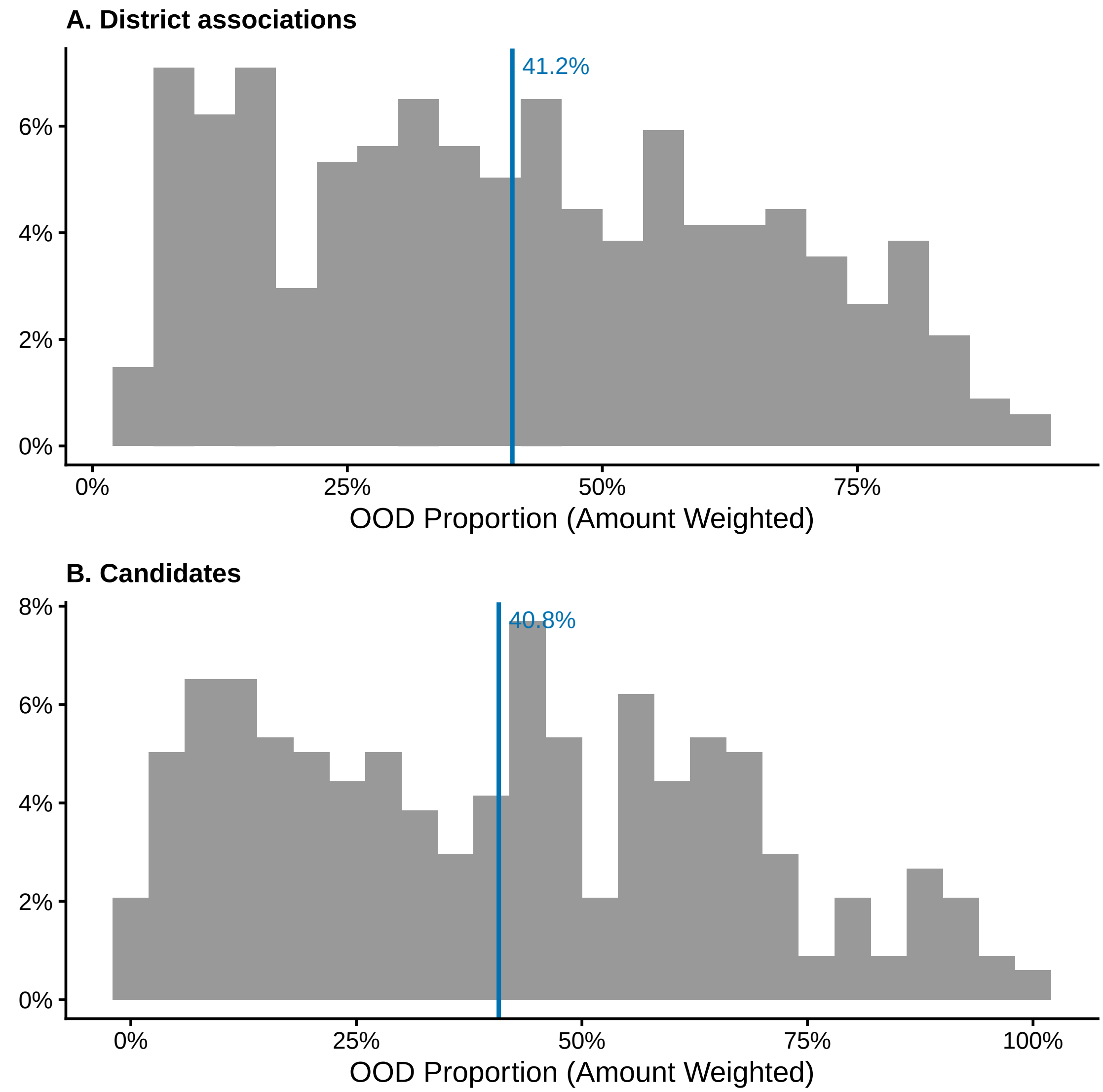


Figure F4: Distribution of out-of-district (OOD) contribution shares for each district in our study (for contributions over $200, from 2015 to 2024). (A) and (B) show this distribution for donations to district associations and candidates separately. The district-level mean is indicated with an orange vertical line in each case. Distributions were similar when raw instead of amount-weighted proportions were used.

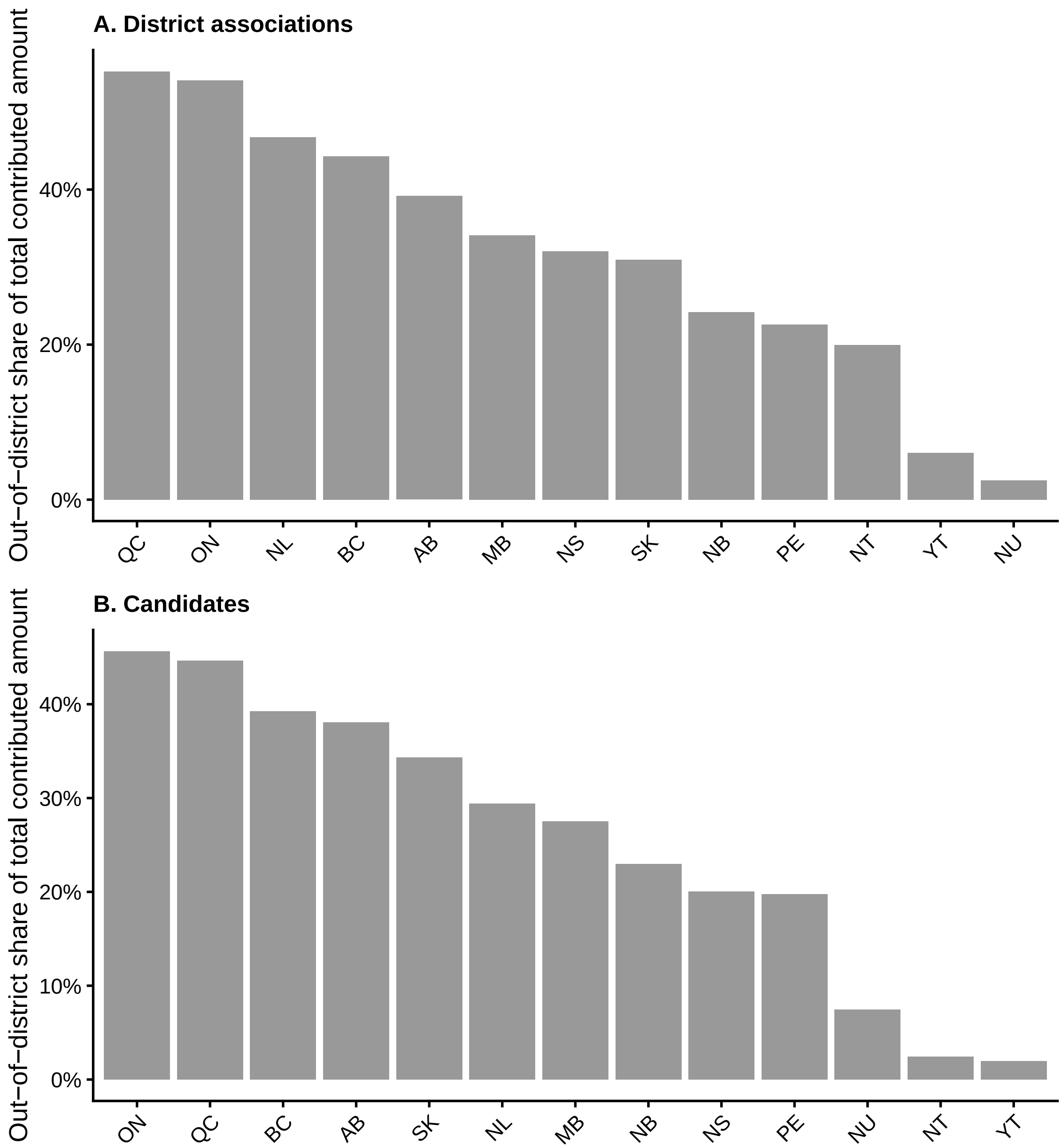


Figure F5: Out-of-district share of the total amount contributed over $200 to (A) district associations and (B) candidates, for each donor province, from 2015 to 2024. Within each panel, provinces are ordered from highest to lowest out-of-district share.

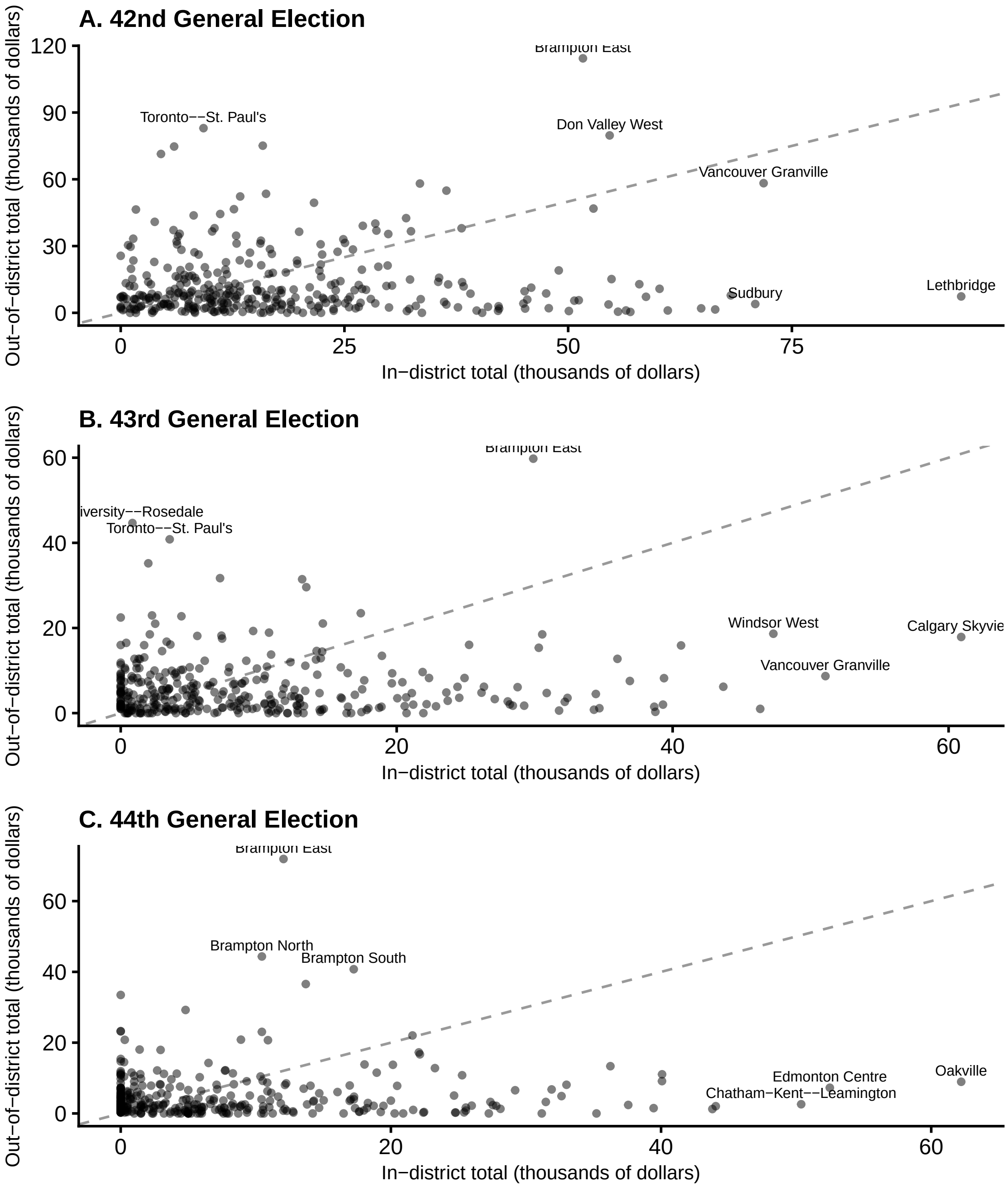
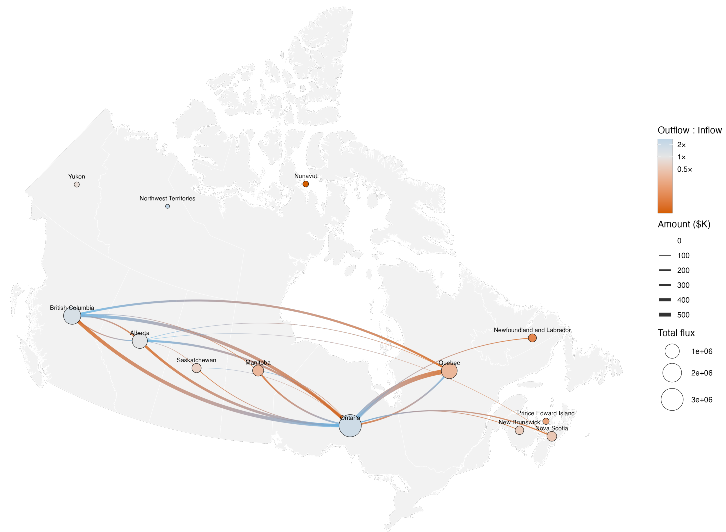


Figure F6: Relationship between the sum of contributions over \$200 donated in- v.s. out-of-district, for each district during the 2013 representational order (effective between 2015 and 2024). (A), (B), and (C) display this relationship for donations to candidates during the 42nd, 43rd, and 44th writ periods respectively. Points represent individual districts. The grey dashed line is 1:1. The top 3 districts by amount contributed in- and out-of-district respectively are labelled with their names.

A. District associations



B. Candidates

Figure F7: Interprovincial funding network of contributions over $200 to localized political entities from 2015 and 2024. (A) displays donations to district associations, and (B) displays donations to candidates (42nd, 43rd, and 44th general election periods combined). Node size reflects total interprovincial flux (amount sent + amount received), and edges are weighted by total donation amount. Provinces with red-coloured nodes receive more contributions than they send, and vice-versa for blue nodes. Edges are coloured on a gradient from blue (origin) to red (destination). Networks were similar when counts were displayed instead of amounts.

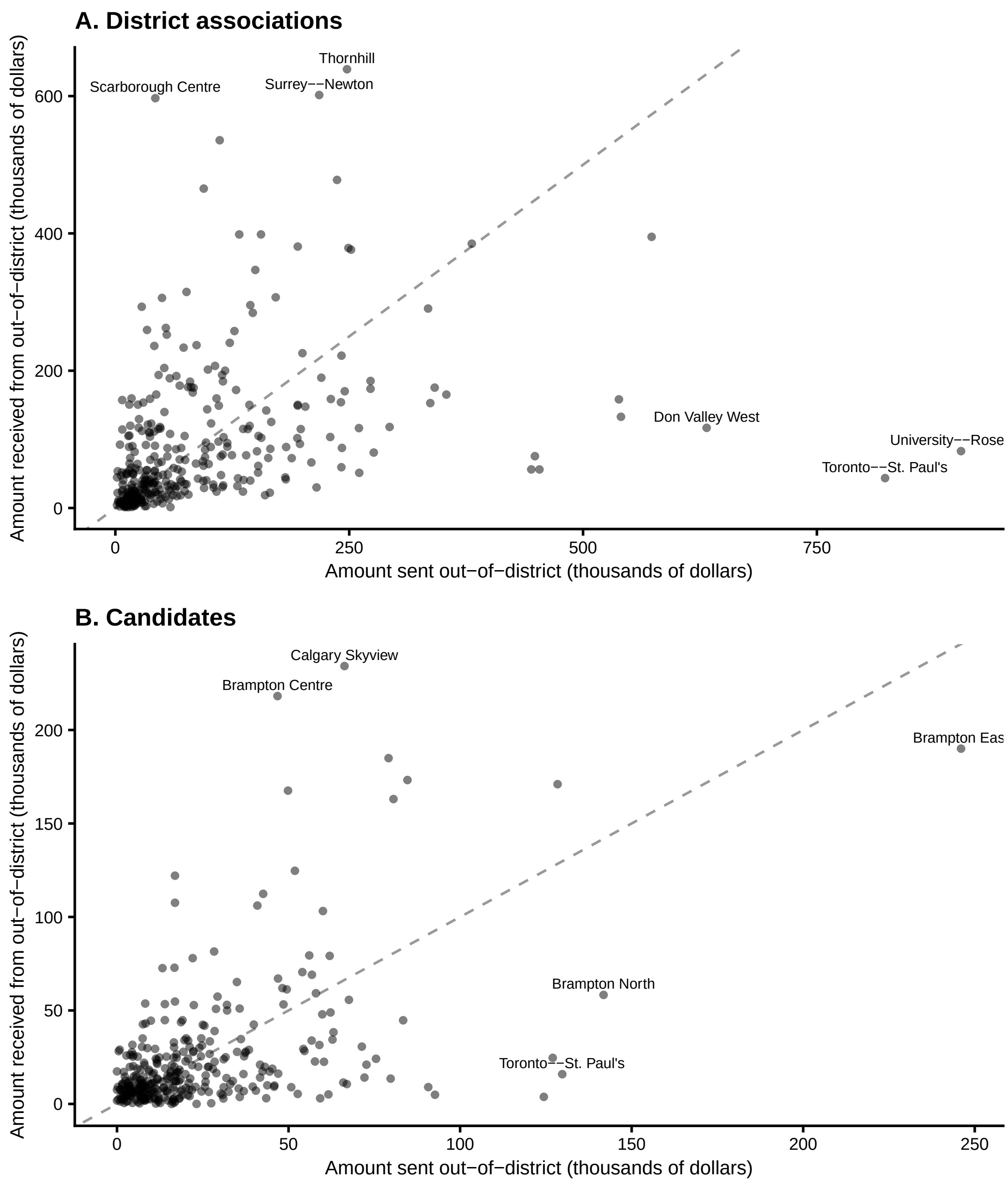


Figure F8: Relationship between the sum of contributions over $200 sent out-of-district v.s. received from out-of-district, for each district during the 2013 representational order (effective between 2015 and 2024). (A) shows this relationship for donations to EDAs, and (B) shows this for donations to candidates. Points represent individual districts. The grey dashed line is 1:1. The top three districts by amount sent and received respectively are labelled with their names. Graphs are similar when counts are displayed instead of amounts. (A) is similar when only donations sent during writ periods are considered, and (B) is similar when specific election cycles are considered.

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